Separating Sampling Error from Conditional Incompatibility in Generative Models

Anonymous Authors[1]

Abstract

Any-to-any generators expose many conditionals of an implied joint, and users compose them by generating an intermediate modality and then the target. If the conditionals are coherent, direct generation $q(z \mid x)$ and intermediate-then-target generation $\int q(z \mid x, y)\, q(y \mid x)\, dy$ agree in distribution. This marginalization-consistency condition is known, and it has been measured without sampling for models with explicit densities. For generators that expose only samples, a measured direct-versus-sequential gap also contains Monte Carlo error, solver discretization, initialization mismatch, guidance and evaluation-design errors. We give a conditional sampling design with the conditioning example as the independent unit, together with a within-example permutation test. We then separate these components on probes whose laws are known in closed form. On a Gaussian probe, guidance with exact scores alone creates population gaps up to 1.909 (standardized), against a Monte Carlo floor of 0.0082. Moderate solver error, by contrast, cancels between the two routes although the sampler is biased. Our first calibration gate failed, and the next one was found defective before use. The test then passed a pre-registered calibration once. On a Gaussian-mixture probe, a fitted normalized density, coherent by construction, was not flagged ($p = 0.460$), although its distance to the truth is clearly nonzero. On the same probe, flow-matching learners with separate networks were flagged at the target level in three of four training replicates, and a shared network in none; the example-paired difference of gap estimates, averaged over the four fit-pairs, is $-9.70[-12.57, -6.83] \times 10^{-3}$ in favour of the shared network, but one replicate reverses the sign, the shared network is flagged at the joint level in two of four, and its direct conditional is often less accurate. Euler with 128, 256 and 512 steps on shared noise moves these estimates by less than 10^{-3}. No real multimodal data are used.

1. Introduction

Any-to-any generators such as UniDiffuser (Bao et al., 2023), Multi-modal Latent Diffusion (Bounoua et al., 2024), OmniFlow (Li et al., 2025), 4M (Mizrahi et al., 2023), CoDi (Tang et al., 2023) and MODUS (Ye et al., 2026) expose conditionals for many subsets of modalities. These conditionals are routinely composed. UniDiffuser produces image variations by generating a caption and then an image from it, and runs blocked Gibbs sampling between modalities (Bao et al., 2023, Sec. 6.3); 4M and MODUS generate chains of intermediate modalities (Mizrahi et al., 2023; Ye et al., 2026). If all conditionals were induced by one joint distribution, the route would not matter in distribution: for an observation x, an intermediate y and a target z,

$$q_{\mathrm{d}}(z \mid x) \;=\; \int q(z \mid x, y)\, q(y \mid x)\, dy \;=:\; q_{\mathrm{s}}(z \mid x). \quad (1)$$

Equation (1) is the law of total probability. Whether a family of conditionals is compatible with some joint is a classical question (Arnold & Press, 1989). For learned predictors, Klötergens et al. (2026) call (1) *marginalization consistency* and show that tabular foundation models violate it. None of this is our contribution.

What changes for diffusion and flow generators is *how* (1) can be checked. Klötergens et al. (2026) evaluate total variation from the models' predictive heads by exact sums or deterministic quadrature, with "no sampling anywhere in the pipeline". A diffusion model exposes neither density, only a sampler. A sampled discrepancy between the two routes then contains, besides any incompatibility of the learned conditionals, five other things: (i) Monte Carlo error; (ii) discretization error of the ODE or SDE solver; (iii) mismatch between the initial distribution and the noised data law; (iv) classifier-free guidance, which does not sample the intended conditional (Chidambaram et al., 2024; Bradley & Nakkiran, 2024); (v) errors of evaluation design, such as feeding observed rather than generated intermediates,

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

pairing samples by index, or counting generated samples as independent units. Unless these are separated, a measured gap says little about the model.

This paper is about that separation, on synthetic probes only. We make three contributions.

- **A conditional sampling design and its estimands** (Sections 3 and 4). The conditioning example is the independent unit; intermediates are fresh for every sample, branches use independent randomness, and a permutation test relabels samples only within an example. We state what each estimand identifies (the target law, or eight projections of the joint) and keep apart non-detection, distributional correctness and the existence of a coherent joint.
- **An error decomposition on closed-form probes** (Sections 5.2 and 5.3). With exact scores, the output law of a guided probability-flow ODE is available in closed form, so Monte Carlo error, discretization, prior mismatch and guidance can each be isolated; on this probe the test passed a pre-registered calibration once, after failures kept on record.
- **Coherent-joint controls** (Section 5.4). Exact and fitted Gaussian mixtures are normalized densities, so their conditionals are coherent by construction while their fit quality varies. This separates coherence from correctness in a learned model.
- **A neural pilot with four training replicates** (Sections 5.5 and 5.6). Separately trained conditional flow learners are flagged at the target level in three of four replicates; a shared network is flagged there in none but at the joint level in two; the arms are compared by an example-paired difference, conditional on the four fits, whose sign reverses in one replicate.

The central comparison, separately learned conditionals against a shared conditional network, is run as a synthetic pilot with four training replicates (Section 5.6); its scope is one probe. We do not claim as new: the identity (1); the consistency and compatibility definitions; the fact that marginal-level consistency does not imply joint-level consistency, which is Proposition 2 of Klötergens et al. (2026); guidance distortion; permutation-test validity (Ernst, 2004; Lehmann & Romano, 2005; Phipson & Smyth, 2010); energy statistics (Székely & Rizzo, 2013; Baringhaus & Franz, 2004).

2. Related Work

Compatibility of conditionals. Arnold & Press (1989) characterize when conditionals admit a joint distribution. Compatible conditionals may still imply an improper joint (Hobert & Casella, 1998). Pseudo-Gibbs sampling with incompatible conditionals has a scan-order-dependent limit (Heckerman et al., 2000; Chen & Ip, 2015; Liu et al., 2014). We use only the definition, and ask how to measure one compatibility relation from samples.

Self-consistency of learned models. Masked language models produce conditionals that no single joint can explain (Young et al., 2022); any-order autoregressive models define one distribution in redundant ways (Shih et al., 2022). Generative marginalization models add a squared log-space penalty on single-step marginalization constraints for discrete data with explicit likelihoods (Liu et al., 2024). This is the closest prior training-time consistency loss. For diffusion language models, Kim (2026) relate order dependence to incompatible denoising conditionals. Closest to our estimand, Klötergens et al. (2026) define marginalization consistency (C1) and factorization consistency (C2) and prove that C2 implies C1 but not conversely. They measure both by total variation from the explicit predictive heads of tabular foundation models and find violations for every model and dataset. We adopt their C1 as our target-level estimand. We add nothing to the definitions; our setting differs only in that the generator exposes samples.

Any-to-any generation and coherence. UniDiffuser trains one noise-prediction network with independent per-modality timesteps; a maximal timestep yields a marginal, which also serves as the unconditional model for guidance (Bao et al., 2023). Multi-modal Latent Diffusion trains a masked multi-time score network on the latents of frozen unimodal autoencoders (Bounoua et al., 2024), and measures coherence with pre-trained classifiers, as in the multimodal VAE literature (Wu & Goodman, 2018; Shi et al., 2019; Sutter et al., 2021; Daunhawer et al., 2022). Both are *shared conditional networks*: one network serves many conditionals. That by itself does not make them the conditionals of one normalized joint. Chung et al. (2025) evaluate cyclic consistency and equivariance of any-to-any models with per-sample criteria. None of these works tests (1) as an equality of distributions.

Guidance, solvers and testing. Classifier-free guidance (Ho & Salimans, 2022) does not sample the guided target it is often described by (Chidambaram et al., 2024; Bradley & Nakkiran, 2024). We therefore treat it as a confound to be measured, not as model error. We use the probability-flow ODE (Song et al., 2021) with the step schedule of Karras et al. (2022). We use energy distances (Székely & Rizzo, 2013; Baringhaus & Franz, 2004) (kernel tests (Gretton et al., 2012) would serve equally well), permutation p-values with the $+1$ correction (Phipson & Smyth, 2010; Ernst, 2004) and the CRPS as a strictly proper

score (Gneiting & Raftery, 2007).

3. Problem Setup

A generator provides samplers for $q_{\mathrm{d}}(z \mid x)$, $q(y \mid x)$ and $q(z \mid x, y)$, and possibly a direct joint sampler $q_{\mathrm{d}}(y, z \mid x)$. Here x is observed, y is an intermediate modality and z is the target. Conditioning examples $x_1, \ldots, x_N$ are drawn i.i.d. from a test split.

Definition 3.1 (Target- and joint-level gaps)**.** For a distance D between distributions, the *target-level gap* is $\Delta_T(x) = D\big(q_{\mathrm{d}}(\cdot \mid x),\, q_{\mathrm{s}}(\cdot \mid x)\big)$ with q_{s} as in (1). When a direct joint sampler exists, the *joint-level gap* is $\Delta_J(x) = D\big(q_{\mathrm{d}}(\cdot, \cdot \mid x),\, q(y \mid x)\, q(z \mid x, y)\big)$.

$\Delta_T \equiv 0$ is C1 of Klötergens et al. (2026). Δ_J compares the sequential pair with a direct joint sampler. It is not their factorization consistency (C2), which compares the two chain-rule orders; we do not measure C2.

In practice we estimate a *projected* joint gap. For a fixed set U of eight unit directions,

$$\Delta_J^U(x) = \frac{1}{|U|} \sum_{u \in U} D\big(\pi_u q_{\mathrm{d}}(\cdot, \cdot \mid x),\, \pi_u\, q_{\mathrm{s}}(\cdot, \cdot \mid x)\big), \quad (2)$$

where π_u maps a law on (y, z) to the law of $u^\top(y, z)$ and $q_{\mathrm{s}}(y, z \mid x) = q(y \mid x) q(z \mid x, y)$. $\Delta_J^U = 0$ is implied by joint equality but does not imply it: two joints can agree on finitely many projections and differ elsewhere. Within the Gaussian family used below, eight directions identify the law; in general they do not.

Three different claims. We keep three statements apart; Table 17 (appendix) lists what would support each. *Non-detection*: a test at a given design (N, M, B) did not reject. *Distributional correctness*: a branch samples the law it is meant to sample, e.g. the true conditional. *Coherent joint existence*: a single joint distribution induces every conditional the model exposes. Non-detection is bounded by power and resolution and does not establish $\Delta_T = 0$. Even $\Delta_T = 0$ does not establish correctness, as Proposition 3.4 and the common-mode solver bias of Section 5.2 show. Equality of one relation is necessary for a coherent joint, but it is not sufficient. This paper measures the first kind of statement for two relations, Δ_T and Δ_J^U, together with quality metrics that bear on the second. The third holds by construction only for the coherent controls of Remark 3.6; it is measured for no model.

Neither check establishes a global joint for all conditionals of an any-to-any model. Each tests one relation.

Proposition 3.2 (Target-level equality is compatibility of the triple)**.** *There is a joint $r(y, z \mid x)$ whose $y \mid x$ marginal is $q(y \mid x)$, whose $z \mid x, y$ conditional is $q(z \mid x, y)$, and whose $z \mid x$ marginal is $q_{\mathrm{d}}(z \mid x)$, if and only if $q_{\mathrm{s}}(\cdot \mid x) = q_{\mathrm{d}}(\cdot \mid x)$.*

The proof is two lines (Section A). Because the first two properties force $r = q(y \mid x)\, q(z \mid x, y)$, target-level equality does not constrain the model's *direct joint* sampler.

Remark 3.3 (Target level does not determine the joint). In a Gaussian model, replace the coefficient b of $y - m_{y|x}(x)$ in $\mathbb{E}[z \mid x, y]$ by cb and the residual variance by $\tau^2 - c^2 b^2 v_y$, where $\tau^2 = \mathrm{Var}(z \mid x)$ and $v_y = \mathrm{Var}(y \mid x)$. Then $q_{\mathrm{s}}(z \mid x) = p(z \mid x)$ for every c, while $\mathrm{Cov}(y, z \mid x)$ changes from bv_y to cbv_y. This is a continuous instance of the separation in Proposition 2 of Klötergens et al. (2026), not a new result. We use it as a control that a target-level check must *not* detect.

Proposition 3.4 (Collapse passes consistency)**.** *If every branch returns the conditional mean deterministically, then q_{d} and q_{s} are the same point mass and $\Delta_T = \Delta_J = 0$. For a Gaussian truth $\mathcal{N}(m, \sigma^2)$, the expected CRPS of that point mass is $\sigma\sqrt{2/\pi}$, against $\sigma/\sqrt{\pi}$ for the true conditional: $\sqrt{2} - 1 \approx 41.4\%$ worse.*

Proposition 3.5 (Observed intermediates leak)**.** *Let $(y^{\mathrm{obs}}, z^{\mathrm{obs}})$ be drawn jointly given x, and let S be a strictly proper, negatively oriented score. Then $\mathbb{E}\, S\big(p(\cdot \mid x, y^{\mathrm{obs}}), z^{\mathrm{obs}}\big) \le \mathbb{E}\, S\big(p(\cdot \mid x), z^{\mathrm{obs}}\big)$.*

This follows from propriety applied conditionally on (x, y^{obs}) (Section A). So a "sequential" branch fed with the observed intermediate scores better than the direct branch while answering a different question. It must not be used to test (1). Chain generation $z \sim q(z \mid y)$, which drops x at the second stage, agrees with $p(z \mid x)$ only if $z \perp x \mid y$. A mismatch there is expected and is not evidence of incompatibility.

Remark 3.6 (Normalized joints are coherent controls). If all conditionals of a model are computed from one normalized density $r(x, y, z)$, then $\Delta_T = \Delta_J = 0$ for every x with $r(x) > 0$ (Proposition 3.2 with that r). For a Gaussian mixture, the weights of $r(z \mid x, y)$ times $r(y \mid x)$ reduce to $\pi_k(x)\, \mathcal{N}_k(y \mid x)$, so the integral in (1) is available component-wise (Section A). Such a model can fit the data badly and still be exactly coherent. We call it a JOINT_COHERENT_CONTROL. A network shared across conditionals, a SHARED_CONDITIONAL_NET, carries no such guarantee.

4. Sampling Design, Tests and Error Decomposition

4.1. Conditional sampling design

For each conditioning example i we draw A_i, M i.i.d. samples from the direct branch, and C_i, M i.i.d. samples from the sequential branch. Every sequential sam-

ple uses a fresh intermediate $\hat{y}_{ij} \sim q(y \mid x_i)$ followed by $z_{ij} \sim q(z \mid x_i, \hat{y}_{ij})$. The first design (v1) also draws a second direct set B_i to measure the Monte Carlo floor. For joint-level tests, a sample is the vector $(\hat{y}_{ij}, z_{ij})$. The assumptions behind every test below are:

Assumption 4.1. (A1) Examples are independent. (A2) Within each example and branch, samples are i.i.d. given x_i. (A3) The branches of an example are generated with independent randomness.

Three common shortcuts break these assumptions. Generating several targets from one intermediate draw violates (A2): target samples then share a latent and are not exchangeable with direct samples. Reusing initial noise across branches, i.e. common random numbers, violates (A3). Pooling or relabelling samples across examples mixes different null distributions. Generated samples, sign flips and permutations are never counted as independent units; only the N examples are.

4.2. Energy-distance estimators

For samples $a_{1:n}$, $c_{1:m}$, the V-statistic energy distance is $\mathrm{ED}_V = \frac{2}{nm}\sum_{j,k}|a_j - c_k| - \frac{1}{n^2}\sum_{j,k}|a_j - a_k| - \frac{1}{m^2}\sum_{j,k}|c_j - c_k|$. The U-statistic ED_U divides the within-sample sums by $n(n-1)$ and $m(m-1)$ instead. In one dimension both are computed exactly in $O((n+m)\log(n+m))$ time from sorted samples.

Lemma 4.2 (standard)**.** *If a is drawn i.i.d. from P and c i.i.d. from Q, then $\mathbb{E}\,\mathrm{ED}_U = D(P,Q)$ and $\mathbb{E}\,\mathrm{ED}_V = D(P,Q) + \mathbb{E}_P|X - X'|/n + \mathbb{E}_Q|Y - Y'|/m$, where D is the population energy distance.*

The v1 checker used the noise-floor-subtracted contrast $d_i = \mathrm{ED}_V(A_i, C_i) - \mathrm{ED}_V(A_i, B_i)$. By Lemma 4.2, $\mathbb{E}d_i = D + (\mathbb{E}|C - C'| - \mathbb{E}|B - B'|)/M$. This is exactly 0 under H_0 and biased under alternatives whose spread differs from the reference; the bias is negative for under-dispersed C. $\mathrm{ED}_U(A_i, C_i)$ is unbiased for D and needs no third set. Being unbiased for a non-negative quantity, ED_U can be negative; its intervals are unconstrained normal approximations over examples, and a negative interval is not a claim of a negative population distance (implementation re-checked on stored samples, Section K). We report effect sizes on a scale standardized by the training-split standard deviation of each coordinate.

4.3. Two randomization tests

Test S (v1): sign flips of d_i. Under H_0 and Assumption 4.1, B_i and C_i are exchangeable given A_i, so d_i is symmetric about 0 and independent across i. The one-sided sign-flip test of $\sum_i d_i$ therefore has level at most α. This is standard randomization logic. Each flip swaps the whole sets B_i and C_i, so there are only two allocations per example and the noise of B_i enters d_i.

Test P (revised): relabelling within examples. Under H_0 and Assumption 4.1, the $2M$ pooled samples of example i are exchangeable. We reassign branch labels uniformly over all $\binom{2M}{M}$ allocations independently in every example, never across examples, and relabel vectors as units. The pre-specified statistic is $T = \frac{1}{N}\sum_i \mathrm{ED}_V(a_i, c_i)$ in the upper tail. The Monte Carlo p-value is $(1 + \#\{b : T_b \geq T_{\mathrm{obs}}\})/(B+1)$, with ties counted as extreme, so it is never zero (Phipson & Smyth, 2010). On small designs the product group is enumerated exactly.

Proposition 4.3 (V and U give the same permutation test)**.** *With equal group sizes, the permutation p-values of T built from ED_V and from ED_U coincide.*

Given the pooled multiset of an example, the within- and cross-sums add to a constant, so both statistics are increasing affine functions of the cross-sum with slopes depending only on M (Section A). The choice between V and U thus matters for effect sizes, not for the test.

What Test P can and cannot establish. For the target level, Test P has level at most α under H_0^T: $q_{\mathrm{d}}(z \mid x_i) = q_{\mathrm{s}}(z \mid x_i)$ for all i. For the joint level, the pooled vectors are exchangeable only under equality of the *full* joint laws, so the level guarantee holds under that null. Its statistic depends only on the projections in U, so it has power only against differences visible in them (Equation (2)). A rejection therefore shows that the joint laws differ. A non-rejection says nothing about directions outside U.

Resolution and multiplicity. With B permutations the p-value lies on a grid of step $1/(B+1)$; for $B = 199$ the smallest value is 0.005. Several conditions together make $P(p \leq \alpha) \leq \alpha$ under H_0 (Phipson & Smyth, 2010):

- the observed allocation is counted (+1) and ties are counted as extreme;
- the B allocations are drawn i.i.d. and uniformly from the within-example relabelling group;
- the statistic is fixed in advance.

Our implementation meets all three. Equality $P(p \leq \alpha) = \lfloor \alpha(B+1) \rfloor/(B+1)$ needs in addition that the observed and permuted statistics have no ties almost surely. That fails in general: the group is finite, ED_V is invariant under complementing an allocation, and collapsed samples tie. We therefore claim level $\leq \alpha$, not exact size. A Bonferroni family of m tests can reject at all only if $m \leq \alpha(B+1) = 10$.

4.4. Error decomposition on a Gaussian probe

Energy distances do not add, so we define components operationally: each is the gap that remains when all other sources are switched off in a probe where every law is computable.

- Monte Carlo error vanishes as $M \to \infty$ (Lemma 4.2).
- Discretization vanishes as the number of solver steps $N_s \to \infty$.
- Prior mismatch is the residual gap of the exact flow started from $\mathcal{N}(0, \sigma_{\max}^2)$ rather than from the noised data law.
- Guidance vanishes at $w = 0$.
- Model incompatibility is whatever remains in a learned model.

Proposition 4.4 (Guided PF-ODE on Gaussian conditionals is affine)**.** *Let the target be $\mathcal{N}(m_c, v_c)$ and the unconditional law $\mathcal{N}(m_u, v_u)$. Use exact noised scores in the variance-exploding parameterization, guided score $(1 + w)s_c - ws_u$, initialization $x_0 \sim \mathcal{N}(0, \sigma_{\max}^2)$, and Euler or Heun steps on any schedule. Then the output is $x_N = g\, x_0 + h_c m_c + h_u m_u$, with (g, h_c, h_u) depending only on (v_c, v_u, w) and the schedule.*

The velocity $\sigma\big[(1 + w)\frac{x-m_c}{v_c+\sigma^2} - w\frac{x-m_u}{v_u+\sigma^2}\big]$ is affine in (x, m_c, m_u), and so is each Euler or Heun step. Composing the two stages of the sequential branch keeps the output Gaussian, so both branch laws are available per example in closed form. We verify this against the sampler to 10^{-8} in unit tests; that is a numerical check, and the proof is the affinity argument itself. The continuous-time limit is also closed-form: the exact flow keeps $(x - m)/\sqrt{v + \sigma^2}$ invariant.

4.5. Quality metrics

Consistency is always reported next to quality, because Proposition 3.4 shows that consistency can be satisfied by a useless generator.

- **Quality to truth.** Where the truth is known, we report the population energy distance between the model's conditional and the true one. For a one-dimensional law, half of it is the CRPS divergence (Gneiting & Raftery, 2007).
- **Proper score.** Otherwise we use the fair ensemble CRPS against held-out data.
- **Diversity.** The sample standard deviation over the true conditional one.

These answer a different question from the compatibility test. Their detection rates are not comparable to its power. Same-index paired MSE between branches is $2\,\mathrm{Var}$ for independent exact samplers and 0 for collapsed ones, so it is not a compatibility metric.

5. Experiments

5.1. Probes and protocol

Two synthetic probes are used; neither is real multimodal data.

- **Gaussian probe.** Three scalars (x, y, z) with a non-Markov and a Markov correlation structure (Section B). Its laws are closed-form for exact samplers and for the PF-ODE.
- **Mixture probe.** A fixed three-component Gaussian mixture over (x, y, z) with multimodal, non-Gaussian conditionals (Section J).

In both, splits are independent draws. The train split is used for fitting and standardization, the dev split for pipeline smoke runs and, for the neural arms, for the pre-registered diagnostics and labels, which select nothing, and the test split ($N = 200$ conditioning examples) for reported results. Model, data and sampling seeds are separate, and every configuration was committed before its test run (Section M); closed-form curves were computed after the runs and are exploratory. The v1 Gaussian-probe runs used Test S with $M = 256$ and 999 flips (controls in Section H); the later runs use Test P with $N = 200$, $M = 64$ and $B = 199$.

5.2. Where sampled gaps come from

Monte Carlo. The floor $\overline{\mathrm{ED}_V(A, B)}$ follows $2\,\mathbb{E}|X - X'|/M$ from $M = 32$ to 512 (Figure 1a). A 0.1τ mean shift is detected at every M in that range with $N = 200$. At $M = 512$, $\bar{d}$ is 5.1×10^{-3} against a population value of 5.6×10^{-3}.

Discretization and prior mismatch. With exact scores and $w = 0$, the Euler direct-vs-sequential gap is 0.00247 at 4 steps and is detected ($p = 0.014$, v1 Test S). From about 6 steps on it is below the one-sided detection resolution of this design, 0.00143 (Figure 1b; Table 5 in Section E). It plateaus at 0.00017 at 64 steps, which equals the closed-form continuous-time limit, 0.00017. The plateau is therefore prior mismatch at $\sigma_{\max} = 80$, not discretization. Over the same range the direct sampler itself stays biased relative to the exact law. At 16 steps its gap to the exact law is 0.0095, which is detected, and its sd ratio is 0.83. Solver error is thus largely common to the two branches and cancels in the comparison: *passing the consistency check*

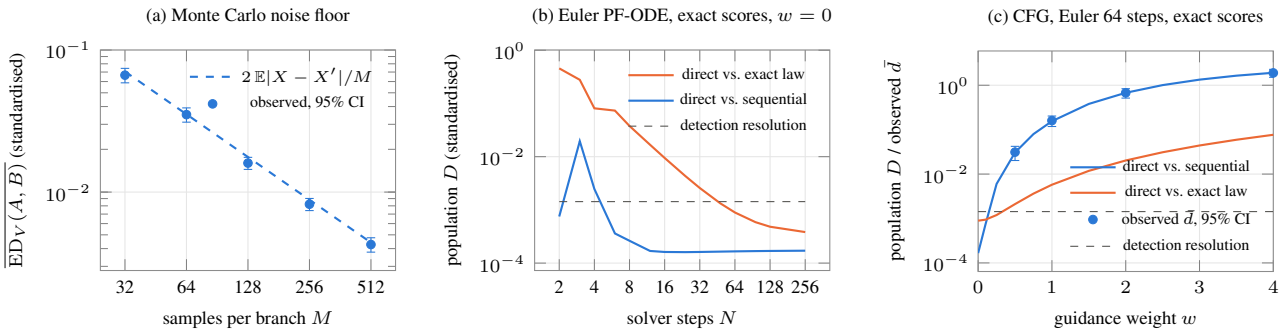


Figure 1. Separating numerical components on the Gaussian probe. (a) The Monte Carlo floor of the V-statistic energy distance between two independent sets follows $2\,\mathbb{E}|X-X'|/M$ (P5-E5, $N=200$). (b) With exact scores and no guidance, the closed-form gap between the direct and sequential PF-ODE laws falls below the one-sided 5% detection resolution of the v1 design ($N=100$, $M=256$) from about 6 steps on, and plateaus at the prior-mismatch value, although the sampler remains biased relative to the exact law. (c) Guidance alone creates direct-vs-sequential gaps far above the resolution; the observed means (P5-E7) match the closed form. Closed-form curves are computed from the affine output law (Proposition 4.4); no Monte Carlo is involved in them.

does not certify a correct sampler, and "not detected" is not "zero". Heun with 4 steps is unstable on this schedule (sd ratio 4.35).

Guidance. Guidance alone produces population gaps of 0.031, 0.155, 0.677 and 1.909 at $w=0.5, 1, 2, 4$, against a floor of about 0.008 (Figure 1c). The observed means agree with the closed form. The guided direct sampler stays close to the exact law: $D=0.0058$ at $w=1$ and 0.077 at $w=4$. Because $\sqrt{D}$ is a metric (Székely & Rizzo, 2013), most of the direct-vs-sequential gap must come from the sequential branch, which applies guidance at both stages. Under our pre-registered decision rule, a direct-vs-sequential gap measured with guidance on is not attributed to the model. This agrees with prior analyses of guidance (Chidambaram et al., 2024; Bradley & Nakkiran, 2024). The magnitudes are specific to this probe.

5.3. A calibrated permutation test on the Gaussian probe

Test P passed one pre-registered calibration (Table 9; $N=200$, $M=64$, $B=199$; 19.9 minutes on two processes).

- **Primary exact null.** 16/400 rejections, two-sided 95% Clopper–Pearson interval [0.023, 0.064]. The gate's one-sided 95% upper bound was $0.060 \le 0.10$.
- **Other null cells.** Three cells showed no excess at the Bonferroni level: the exact null at the joint level, a correlation flip that preserves the target law, and a Markov chain.
- **Alternatives.** Four fixed alternatives were all powered: 45/50, 26/50, 50/50 and 50/50.
- **Quality control.** A both-branch collapse is a compatibility null and a quality failure. It was never rejected, and the quality metrics flagged it in every replicate.

This is engineering evidence for this implementation and design. The level-α guarantee comes from exchangeability and a fixed statistic, not from this run. It transfers to a new model only if Assumption 4.1 holds for that model's sampling pipeline. The secondary null cells detect a true size of 0.10 with probability only 0.42. The route to this result included a failed gate ($p=0.026$ on a single exact-null test, traced to the gate design) and a gate withdrawn before use (it would have failed a valid test with probability 0.88); both are kept on record with the full table in Section I (Table 9).

5.4. Coherent-joint controls on the mixture probe

Two models are compared, each a normalized density whose conditionals are coherent by Remark 3.6.

- TRUE_JOINT_CONTROL: the exact source mixture.
- FITTED_JOINT_CONTROL: a two-component mixture fitted by EM on the training split (37 iterations). It is misspecified by design, since the source has three components.

All conditionals of each model come from its own density and are sampled exactly, so no solver is involved. The primary endpoint is the target test. The projected-joint test is secondary. Table 1 reports the single pre-registered run (22.5 s).

Coherence. Neither control is flagged by the primary target test: $p=0.075$ and 0.460, Holm-adjusted over the primary family 0.15 and 0.46. The gap estimates contain 0. The exact mixture's secondary projected-joint test gave $p=0.035$, with gap estimate $2.08[-1.67, 5.82]\times 10^{-3}$.

Table 1. Coherent-joint controls on the mixture probe (Test P, $N = 200$, $M = 64$, $B = 199$). $\widehat{\Delta}_T$ is the mean unbiased energy distance between the two branches. $D(\text{model}, \text{truth})$ is the closed-form energy distance between the model's and the true conditional of z given x, averaged over test examples. Both are standardized.

Model	p_T	p_J	$\widehat{\Delta}_T \times 10^3$ [95% CI]	$D(\text{model, truth}) \times 10^3$ [95% CI]	test log-lik.
True joint (exact mixture)	0.075	0.035	2.71 [-1.38, 6.80]	0.00 [0.00, 0.00]	-3.390
Fitted joint (EM, $K = 2$)	0.460	0.250	0.21 [-3.03, 3.46]	30.10 [25.26, 34.95]	-3.623
Neural flow arms (round 6)	see Table 11				

Its population gap is exactly zero, so this is a false positive of a level-0.05 test, one of four tests run. We report it as observed; it is not evidence of incompatibility. The coherence of both controls follows from their construction (Remark 3.6), not from these p-values; non-rejection is only consistent with it.

Correctness. The fitted control is coherent yet wrong: its conditional of z given x lies at energy distance $0.030[0.025, 0.035]$ from the truth and its projected joint at $0.021[0.018, 0.024]$; its test log-likelihood is -3.623 against -3.390 for the source; its conditional spread is too large by a factor of 1.056. A compatibility test cannot see this by construction; only the quality metrics can. Table 18 (appendix) summarizes what the runs do and do not decide.

5.5. Neural flow learners: development pilot

The central experiment was first run once as a development pilot (CPU only), after the controls of Section 5.4 had been seen; Section 5.6 replicates it over four fixed initializations on a new test set.

Two learners were trained on the probe's training split with linear-path conditional flow matching (Section J): INDEPENDENT_CONDITIONAL_FM, four MLPs with one per conditional (4×5000 updates), and SHARED_CONDITIONAL_NET, one MLP over (x, y, z) whose input marks each coordinate as observed, target or absent (20000 updates); parameter counts are close (34757 and 34819) but the arms are not equal-compute, and the final iterate is the model. Sampling uses Euler from $\mathcal{N}(0, I)$ with guidance off; the sequential branch draws a fresh y from the arm's own $q(y \mid x)$ for every sample. In the pilot (Section K) the independent arm was flagged at the target endpoint ($p = 0.005$, Holm 0.010; gap $11.90[7.19, 16.61] \times 10^{-3}$) and the shared arm was not ($p = 0.425$; gap $0.24[-2.75, 3.24] \times 10^{-3}$), with a paired difference of $-11.65[-16.93, -6.38] \times 10^{-3}$; the shared arm's projected joint was flagged ($p = 0.035$, unadjusted secondary result) and its direct branch was less accurate. It used one seed per arm, shared sampling streams between dev and test, and its dev p-values were seen before the design below was fixed, so it is kept as a development result and not re-scored.

Table 2. Paired between-arm difference at the target level ($\times10^3$): per-replicate means $\bar{d}_r(h)$ with pointwise 95% CIs, and $\widehat{\Theta}_h$ over examples (conditional on the four fits) with the example bootstrap; $h = 256$ lies between the two columns (Table 12, appendix).

Quantity	$h = 128$ (primary)	$h = 512$
$\bar{d}_{R1}(h)$	-11.18 [-17.70, -4.66]	-11.10 [-17.63, -4.58]
$\bar{d}_{R2}(h)$	-22.81 [-29.58, -16.04]	-22.57 [-29.35, -15.80]
$\bar{d}_{R3}(h)$	-8.15 [-13.11, -3.19]	-8.21 [-13.22, -3.21]
$\bar{d}_{R4}(h)$	3.35 [-1.98, 8.67]	3.30 [-2.04, 8.64]
$\widehat{\Theta}_h$ (normal)	-9.70 [-12.57, -6.83]	-9.65 [-12.53, -6.77]
$\widehat{\Theta}_h$ (bootstrap)	[-12.47, -6.89]	[-12.57, -6.82]

5.6. Four training replicates with solver sensitivity

Four fixed training replicates R_1–R_4 were then run once with the same training split, architecture, update budget and optimizer, new integer seeds for initialization and minibatch order, and a new test set of 200 examples from the same source, disjoint from the pilot's. Every draw is keyed by split, replicate, arm, pattern, example, branch, sample and purpose, without the step count, so Euler with 128 (primary), 256 and 512 steps integrates the same initial noise; branches share no stream, and a fresh y is drawn for every z. The eight target tests at 128 steps form one auxiliary family with Holm correction (with $B = 199$ the smallest attainable p is 0.005 against a first threshold of 0.00625); the eight projected-joint tests are a separate secondary family; no permutation test is run at 256 or 512 steps. Per-replicate results are in Table 13 (appendix).

Estimand. For replicate r and example i let $d_{ri}(h) = \widehat{\Delta}_T^{\text{shared}}(h) - \widehat{\Delta}_T^{\text{indep}}(h)$, the difference of the two arms' unbiased gap estimates at step count h; negative values favour the shared arm. The primary quantity is $\Theta_{128} = \text{mean}_i\, \text{mean}_r\, d_{ri}(128)$, a comparison conditional on these four fit-pairs, not an effect over a population of training runs. Intervals are paired at the example (normal over the 200 per-example replicate means; percentile bootstrap with 2000 resamples of examples carrying all replicates, arms and step counts); replicate means are shown side by side and never pooled as 800 values.

Replication. At the primary solver, 3 of the four independent arms and 0 of the four shared arms are flagged after Holm correction (raw p in Table 13). The per-replicate means of $d(128)$ are $-22.81, -11.18, -8.15, 3.35$

($\times 10^{-3}$; between-replicate sd 10.75), and $\widehat{\Theta}_{128} = -9.70[-12.57, -6.83]$ (bootstrap $[-12.47, -6.89]$); sign reversal across replicates: yes. By the pre-registered rule this is MIXED_REPLICATION: three of the four fit-pairs favour the shared arm with pointwise intervals excluding 0, and the fourth reverses the sign with an interval that includes 0. At the joint level, 2 of the four shared arms are flagged in the secondary family, against 4 independent arms; $\widehat{\Theta}^{J}_{128} = -5.51[-7.64, -3.37]$.

Quality and cost. Across replicates the independent arms' direct branches lie 2.4 to 4.6 ($\times 10^{-3}$) from the truth and their sequential routes 2.2 to 14.5; the shared arms' direct branches 2.8 to 9.7 and sequential routes 4.5 to 13.0. This is consistent with error accumulating along the independent arm's two-stage route, but causes were not attributed per route (conditional errors can also cancel). Training took 26 to 29 CPU-s per independent and 37 to 40 per shared arm (Table 15, appendix; not equal-compute); a smaller gap with a worse direct branch is a trade-off, not an improved generator.

Solver sensitivity. Going from 128 to 512 steps changes the gap estimate by a detectable amount in 2 of four independent and 2 of four shared arms (paired intervals, Table 14, appendix), while $\widehat{\Theta}_{512} = -9.65[-12.53, -6.77]$. Agreement between step counts is not a convergence proof, and where paired differences remain detectable the numerical and learned components are not separated.

Equal training CPU (supplementary). Matching update counts gives the shared network about four times the per-update arithmetic, so the eight fits were retrained from step 0 with the same seeds, architectures, loss, optimizer and data, each arm stopped after the first full update at which its process-CPU time reached 60 s (independent: 32,301 to 32,824 updates; shared: 23,058 to 23,730; OS-level, not FLOP, matching), and evaluated on a third test set of 200 examples (Section L). At 128 steps, 2 independent and 1 shared arms are flagged after Holm correction; the per-replicate means of $d(128)$ are $-9.15, -5.97, -4.52, -3.21$ ($\times 10^{-3}$) and $\widehat{\Theta}_{128} = -5.71[-9.09, -2.34]$ (bootstrap $[-9.19, -2.54]$), with no sign reversal but only R_1's pointwise interval excluding 0: again MIXED_REPLICATION. No shared arm is flagged at the joint level after correction, the shared arms' direct branches (3.5 to 7.0, $\times 10^{-3}$) are no longer worse than the independent arms' (3.8 to 13.3); the comparison with the equal-update replicates is descriptive.

6. Limitations

- **Data and models.** All evidence is synthetic: a Gaussian probe, a three-component mixture probe, an EM-fitted mixture and two small flow-matching learners. Nothing here shows that real any-to-any generators are, or are not, incompatible.

- **Closed-form decomposition.** It relies on Gaussian conditionals and exact scores. With learned scores, discretization, prior mismatch and model error interact; the neural gap estimates move with the step count by up to about 10^{-3}, detectable under shared noise but small against the gaps.

- **Tests and conventions.** The joint test compares eight projections and cannot certify joint equality; calibration ran once at one design; the level guarantee needs fresh intermediates and independent branch randomness; non-detection is bounded by power; the unconditional model is the target marginal (Bao et al., 2023); energy distance is one of several discrepancies; C2 is not measured.

- **Pilots.** One data seed, one probe, one fitted control, one pilot and four replicate initializations per arm; the pilot's dev and test stages shared sampling streams (the replicates do not); the arms are not equal-compute (an equal-CPU run is in Section L); the between-arm average is conditional on the four fits; sample-based and closed-form quality estimates share an estimand but not examples.

7. Conclusion

For samplers without densities, checking whether direct and intermediate-then-target generation agree is mostly a matter of accounting for other sources of error. On probes whose laws are known, guidance can dominate the gap, and moderate solver error cancels between routes even when the sampler is wrong. A permutation test with the conditioning example as the unit passed a pre-registered calibration once, after one failed gate and one gate design withdrawn before use. Coherent-by-construction controls are not flagged at the primary endpoint even when their fit is poor (one secondary joint test of the exact control rejected). Consistency is therefore neither correctness nor evidence of a global joint. In a neural pilot replicated over four initializations, separately trained conditionals were flagged in three of four; a shared network had the smaller target gap on average, with one reversal, was flagged at the joint level in two of four, and was often less accurate: sharing changes the gap without defining a joint. At equal training CPU (supplementary) the sign held in all four pairs, detectable in one, and the accuracy disadvantage did not recur. Whether learned any-to-any generators are incompatible beyond these effects remains open.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here. The work concerns evaluation methodology. A test that confuses sampler or guidance artifacts with model incoherence could lead practitioners to wrong conclusions about generated multimodal content. We therefore report our calibration failure and all unrun evidence explicitly. No human or personal data were used.

References

Arnold, B. C. and Press, S. J. Compatible conditional distributions. *Journal of the American Statistical Association*, 84(405):152–156, 1989. doi: 10.1080/01621459.1989.10478750.

Bao, F., Nie, S., Xue, K., Li, C., Pu, S., Wang, Y., Yue, G., Cao, Y., Su, H., and Zhu, J. One transformer fits all distributions in multi-modal diffusion at scale. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 1692–1717. PMLR, 2023.

Baringhaus, L. and Franz, C. On a new multivariate two-sample test. *Journal of Multivariate Analysis*, 88(1): 190–206, 2004. doi: 10.1016/S0047-259X(03)00079-4.

Bounoua, M., Franzese, G., and Michiardi, P. Multi-modal latent diffusion. *Entropy*, 26(4):320, 2024. doi: 10.3390/e26040320.

Bradley, A. and Nakkiran, P. Classifier-free guidance is a predictor-corrector, 2024.

Chen, S.-H. and Ip, E. H. Behaviour of the Gibbs sampler when conditional distributions are potentially incompatible. *Journal of Statistical Computation and Simulation*, 85(16):3266–3275, 2015. doi: 10.1080/00949655.2014.968159.

Chidambaram, M., Gatmiry, K., Chen, S., Lee, H., and Lu, J. What does guidance do? A fine-grained analysis in a simple setting. In *Advances in Neural Information Processing Systems*, volume 37, pp. 84968–85005. Curran Associates, Inc., 2024. doi: 10.52202/079017-2698.

Chung, J., Yoon, J., Park, J., Lee, S., Yang, J., Park, S., and Yu, Y. Are any-to-any models more consistent across modality transfers than specialists? In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 2595–2606, Vienna, Austria, July 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.acl-long.130.

Daunhawer, I., Sutter, T. M., Chin-Cheong, K., Palumbo, E., and Vogt, J. E. On the limitations of multimodal VAEs. In *International Conference on Learning Representations*, 2022.

Ernst, M. D. Permutation methods: A basis for exact inference. *Statistical Science*, 19(4):676–685, 2004. doi: 10.1214/088342304000000396.

Gneiting, T. and Raftery, A. E. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437.

Gretton, A., Borgwardt, K. M., Rasch, M. J., Schölkopf, B., and Smola, A. A kernel two-sample test. *Journal of Machine Learning Research*, 13(25):723–773, 2012.

Heckerman, D., Chickering, D. M., Meek, C., Rounthwaite, R., and Kadie, C. Dependency networks for inference, collaborative filtering, and data visualization. *Journal of Machine Learning Research*, 1:49–75, October 2000.

Ho, J. and Salimans, T. Classifier-free diffusion guidance, 2022.

Hobert, J. P. and Casella, G. Functional compatibility, Markov chains, and Gibbs sampling with improper posteriors. *Journal of Computational and Graphical Statistics*, 7(1):42–60, 1998. doi: 10.1080/10618600.1998.10474760.

Karras, T., Aittala, M., Aila, T., and Laine, S. Elucidating the design space of diffusion-based generative models. In *Advances in Neural Information Processing Systems*, volume 35, pp. 26565–26577. Curran Associates, Inc., 2022. doi: 10.52202/068431-1926.

Kim, J. Path-dependent denoising: A non-conservative field perspective on order collapse in diffusion language models, 2026.

Klötergens, C., Yalavarthi, V. K., Schmidt-Thieme, L., and Hanika, T. Do tabular foundation models agree with themselves?, 2026.

Lehmann, E. L. and Romano, J. P. *Testing Statistical Hypotheses*. Springer Texts in Statistics. Springer, New York, NY, third edition, 2005. ISBN 978-0-387-98864-1. doi: 10.1007/0-387-27605-X.

Li, S., Kallidromitis, K., Gokul, A., Liao, Z., Kato, Y., Kozuka, K., and Grover, A. OmniFlow: Any-to-any generation with multi-modal rectified flows. In *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 13178–13188. IEEE, 2025. doi: 10.1109/CVPR52734.2025.01230.

Liu, J., Gelman, A., Hill, J., Su, Y.-S., and Kropko, J. On the stationary distribution of iterative imputations. *Biometrika*, 101(1):155–173, 2014. doi: 10.1093/biomet/ast044.

Liu, S., Ramadge, P., and Adams, R. P. Generative marginalization models. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 31773–31807. PMLR, 2024.

Mizrahi, D., Bachmann, R., Kar, O. F., Yeo, T., Gao, M., Dehghan, A., and Zamir, A. 4M: Massively multimodal masked modeling. In *Advances in Neural Information Processing Systems*, volume 36, pp. 58363–58408. Curran Associates, Inc., 2023. doi: 10.52202/075280-2544.

Phipson, B. and Smyth, G. K. Permutation P-values should never be zero: Calculating exact P-values when permutations are randomly drawn. *Statistical Applications in Genetics and Molecular Biology*, 9(1):Article 39, 2010. doi: 10.2202/1544-6115.1585.

Shi, Y., Siddharth, N., Paige, B., and Torr, P. H. S. Variational mixture-of-experts autoencoders for multi-modal deep generative models. In *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc., 2019.

Shih, A., Sadigh, D., and Ermon, S. Training and inference on any-order autoregressive models the right way. In *Advances in Neural Information Processing Systems*, volume 35, pp. 2762–2775. Curran Associates, Inc., 2022. doi: 10.52202/068431-0200.

Song, Y., Sohl-Dickstein, J., Kingma, D. P., Kumar, A., Ermon, S., and Poole, B. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations*, 2021.

Sutter, T. M., Daunhawer, I., and Vogt, J. E. Generalized multimodal ELBO. In *International Conference on Learning Representations*, 2021.

Székely, G. J. and Rizzo, M. L. Energy statistics: A class of statistics based on distances. *Journal of Statistical Planning and Inference*, 143(8):1249–1272, 2013. doi: 10.1016/j.jspi.2013.03.018.

Tang, Z., Yang, Z., Zhu, C., Zeng, M., and Bansal, M. Any-to-any generation via composable diffusion. In *Advances in Neural Information Processing Systems*, volume 36, pp. 16083–16099. Curran Associates, Inc., 2023. doi: 10.52202/075280-0707.

Wu, M. and Goodman, N. Multimodal generative models for scalable weakly-supervised learning. In *Advances in Neural Information Processing Systems*, volume 31. Curran Associates, Inc., 2018.

Ye, M., An, Z., Gao, Z., Liu, X., Fleuret, F., Li, C., Zadeh, A., Belongie, S., Dehghan, A., Allardice, J., Mizrahi, D., Kar, O. F., Bachmann, R., and Zamir, A. MODUS: Decoder-only any-to-any modeling of diverse modalities, 2026.

Young, T., Chen, Y., and You, Y. Inconsistencies in masked language models, 2022.

A. Proofs

Proposition 3.2. ($\Leftarrow$) Take $r(y, z \mid x) = q(y \mid x)\, q(z \mid x, y)$. Its $y \mid x$ marginal and $z \mid x, y$ conditional are the given ones, and its $z \mid x$ marginal is $q_{\mathrm{s}} = q_{\mathrm{d}}$. ($\Rightarrow$) Any r with the stated $y \mid x$ marginal and $z \mid x, y$ conditional equals $q(y \mid x)\, q(z \mid x, y)$ wherever $q(y \mid x) > 0$. Its $z \mid x$ marginal is therefore q_{s}, which must equal q_{d}. □

Proposition 3.4. Deterministic branches that return $\mathbb{E}[z \mid x]$, respectively $\mathbb{E}[z \mid x, \mathbb{E}[y \mid x]]$, coincide for Gaussian conditionals, so both laws are $\delta_{\mathbb{E}[z|x]}$. The CRPS of a point mass at m against $Z \sim \mathcal{N}(m, \sigma^2)$ is $\mathbb{E}|Z - m| = \sigma\sqrt{2/\pi}$. The expected CRPS of $\mathcal{N}(m, \sigma^2)$ against its own draw is $\mathbb{E}|Z - m| - \frac{1}{2}\mathbb{E}|Z - Z'| = \sigma\sqrt{2/\pi} - \sigma/\sqrt{\pi} = \sigma/\sqrt{\pi}$. The ratio of the two is $\sqrt{2}$. □

Proposition 3.5. Conditionally on (x, y^{obs}) we have $z^{\mathrm{obs}} \sim p(\cdot \mid x, y^{\mathrm{obs}})$. By propriety, $\mathbb{E}[S(p(\cdot \mid x, y^{\mathrm{obs}}), z^{\mathrm{obs}}) \mid x, y^{\mathrm{obs}}] \le \mathbb{E}[S(Q, z^{\mathrm{obs}}) \mid x, y^{\mathrm{obs}}]$ for every Q, in particular for $Q = p(\cdot \mid x)$. Averaging over (x, y^{obs}) gives the claim. □

Lemma 4.2. The n^2 within-sample terms include n zero diagonal terms, so $\mathbb{E}\frac{1}{n^2}\sum_{j,k}|a_j - a_k| = (1 - \frac{1}{n})\mathbb{E}|X - X'|$, and likewise for c. Hence $\mathbb{E}\,\mathrm{ED}_V = 2\mathbb{E}|X - Y| - (1 - \frac{1}{n})\mathbb{E}|X - X'| - (1 - \frac{1}{m})\mathbb{E}|Y - Y'| = D + \frac{1}{n}\mathbb{E}|X - X'| + \frac{1}{m}\mathbb{E}|Y - Y'|$. The U-statistic removes the diagonal terms. This result is standard (Székely & Rizzo, 2013); we include the derivation because the v1 contrast depends on it. □

Validity of Test S. Under H_0 and Assumption 4.1, (A_i, B_i, C_i) and (A_i, C_i, B_i) have the same distribution, so $d_i \overset{d}{=} -d_i$. The d_i are independent across i, so conditional on $(|d_i|)_i$ the signs are i.i.d. uniform. The Monte Carlo sign-flip p-value with the $+1$ correction is then valid (Phipson & Smyth, 2010). □

Remark 3.6 for Gaussian mixtures. Let $r = \sum_k \pi_k \mathcal{N}_k(x, y, z)$, and write $\pi_k(x) = \pi_k \mathcal{N}_k(x)/r(x)$ and $w_k(x, y) = \pi_k \mathcal{N}_k(x, y)/r(x, y)$. Then $r(y \mid x)\, w_k(x, y) = \frac{r(x,y)}{r(x)} \cdot \frac{\pi_k \mathcal{N}_k(x,y)}{r(x,y)} = \pi_k(x)\, \mathcal{N}_k(y \mid x)$. Hence $\int r(z \mid x, y)\, r(y \mid x)\, dy = \sum_k \pi_k(x) \int \mathcal{N}_k(z \mid x, y)\, \mathcal{N}_k(y \mid x)\, dy = \sum_k \pi_k(x)\, \mathcal{N}_k(z \mid x) = r(z \mid x)$. □A quadrature check of this identity is in the unit tests; it is a numerical check, not part of the proof.

Proposition 4.3. Fix example i and its pooled multiset. Let S_{aa}, S_{cc} and S_{ac} be the within-a, within-c and cross sums of absolute differences over unordered pairs. Then $S_{aa} + S_{cc} + S_{ac} = S_{\mathrm{pool}}$ is invariant under relabelling. With $n = m = M$, $\mathrm{ED}_V = \frac{2S_{ac}}{M^2} - \frac{2(S_{\mathrm{pool}} - S_{ac})}{M^2}$ and $\mathrm{ED}_U = \frac{2S_{ac}}{M^2} - \frac{2(S_{\mathrm{pool}} - S_{ac})}{M(M-1)}$. Both are increasing affine functions of S_{ac}, with slopes that depend only on M and intercepts that are invariant under relabelling. Summing over examples, both versions of T are the same increasing affine function of $\sum_i S_{ac,i}$ up to invariant constants, so they order all allocations identically. □

Proposition 4.4. With $s_c = -(x - m_c)/(v_c + \sigma^2)$ and $s_u = -(x - m_u)/(v_u + \sigma^2)$, the velocity $dx/d\sigma = -\sigma[(1 + w)s_c - w s_u]$ equals $\sigma a(\sigma)x - \sigma[b_c(\sigma)m_c + b_u(\sigma)m_u]$, with $a = \frac{1+w}{v_c+\sigma^2} - \frac{w}{v_u+\sigma^2}$, $b_c = \frac{1+w}{v_c+\sigma^2}$ and $b_u = -\frac{w}{v_u+\sigma^2}$. An Euler step $x \mapsto x + h\, v(x, \sigma)$ and a Heun step are compositions of affine maps in (x, m_c, m_u) with coefficients independent of (x, m_c, m_u). Induction over the steps gives the claim. □

B. Probe Details

- **Conditionals** come from the Schur complement $T \mid G = g \sim \mathcal{N}(\mu_T + \Sigma_{TG}\Sigma_{GG}^{-1}(g - \mu_G),\ \Sigma_{TT} - \Sigma_{TG}\Sigma_{GG}^{-1}\Sigma_{GT})$.
- **Perturbations** act on the second stage $q(z \mid x, y)$:
  - a mean shift adds $\delta\tau$ to the intercept;
  - a variance ratio r sets the residual variance to $r\tau^2 - b^2 v_y$;
  - a correlation scale c follows Remark 3.3;
  - a stage-1 shift adds $\delta\sqrt{v_y}$ to $\mathbb{E}[y \mid x]$.
- **Collapse** sets the sampling temperature to 0.
- **Sampler.** The PF-ODE uses the schedule of Karras et al. (2022) with $\sigma_{\min} = 0.002$, $\sigma_{\max} = 80$, $\rho = 7$, the final step to $\sigma = 0$, and initialization $\mathcal{N}(0, \sigma_{\max}^2)$.

- **Guidance.** The unconditional model for both stages is the marginal of the stage target.
- **Scale.** All discrepancies are computed after dividing y and z by their train-split standard deviations.
- **Sample reuse.** Reference sets are shared across conditions within a run (common random numbers). The Monte Carlo experiment uses nested prefixes across M.

C. Estimator Versus Closed Form for All v1 Cells

For every exact-Gaussian and PF-ODE condition, the closed form of Lemma 4.2 and Proposition 4.4 predicts $\mathbb{E}\bar{d}$ from the laws of the branches alone. Across 43 non-degenerate target-level cells, the observed means lie within $|z| \leq 2.16$ of the prediction, and across 25 joint-level cells within $|z| \leq 2.60$ (Section C). The two target-level cells with $|z| > 2$ are the exact nulls on the two joints, which share their random numbers; see Section I. The spread term of Lemma 4.2 is visible where it should be. For variance ratio 0.8 the population D is 3.3×10^{-3}, while $\mathbb{E}\bar{d} = 2.9 \times 10^{-3}$.

Table 3 lists, for every v1 condition, the population gap, the closed-form expectation of the v1 contrast and the observed mean with its 95% CI. It was computed after the results were seen and is exploratory. z is not defined for the collapsed condition, whose contrasts are identically 0.

Table 3. Closed-form expectation versus observation, all v1 cells (values $\times 10^3$).

Exp.	Condition	M	D_{pop}	$\mathbb{E}[d]$	$\bar{d}$ [95% CI] (all $\times 10^3$)	z	$\mathbb{E}[d_J]$	$\bar{d}_J$
E1-NULL	exact_sequential	256	0.000	0.000	1.23 [0.03, 2.43]	2.00	-0.000	0.64
E2-CHECKER	mean_shift_0.05	256	1.412	1.412	2.03 [0.54, 3.53]	0.81	0.821	1.12
E2-CHECKER	mean_shift_0.1	256	5.646	5.646	4.39 [2.80, 5.98]	-1.55	3.284	3.46
E2-CHECKER	mean_shift_0.2	256	22.558	22.558	22.22 [19.15, 25.29]	-0.22	13.118	13.31
E2-CHECKER	mean_shift_0.5	256	139.768	139.768	136.98 [130.19, 143.78]	-0.80	81.201	81.13
E2-CHECKER	var_ratio_0.8	256	3.321	2.855	2.13 [1.00, 3.25]	-1.27	2.049	2.74
E2-CHECKER	var_ratio_1.1	256	0.657	0.872	0.54 [-0.59, 1.66]	-0.59	0.542	0.97
E2-CHECKER	var_ratio_1.25	256	3.713	4.234	4.81 [3.37, 6.26]	0.79	2.592	2.91
E2-CHECKER	var_ratio_1.5	256	12.793	13.784	13.31 [11.41, 15.21]	-0.49	8.216	8.46
E2-CHECKER	corr_scale_-1	256	0.000	0.000	0.07 [-0.98, 1.12]	0.14	65.473	66.25
E2-CHECKER	corr_scale_0	256	0.000	0.000	-0.16 [-1.37, 1.06]	-0.25	18.050	17.71
E2-CHECKER	corr_scale_0.5	256	0.000	0.000	0.36 [-0.76, 1.49]	0.63	5.615	6.62
E2-CHECKER	stage1_mean_shift_0.5	256	64.684	64.684	64.42 [59.73, 69.11]	-0.11	78.882	80.35
E3-COLLAPSE	collapse_both	256	0.000	0.000	0.00 [0.00, 0.00]	NA	0.000	0.00
E3-COLLAPSE	collapse_intermediate	256	22.789	21.618	22.75 [20.97, 24.52]	1.25	109.412	111.60
E4-INTERMEDIATE	observed_intermediate	256	323.306	322.135	320.78 [269.99, 371.57]	-0.05	–	–
E4-INTERMEDIATE	chain_drop_x	256	40.017	39.742	39.28 [31.13, 47.44]	-0.11	23.491	23.41
E4-MARKOV	exact_sequential	256	-0.000	-0.000	1.20 [0.11, 2.29]	2.16	-0.000	0.58
E4-MARKOV	chain_drop_x	256	-0.000	-0.000	-0.18 [-1.08, 0.71]	-0.40	-0.000	0.19
E6-SOLVER	euler_2_seq	256	0.751	0.778	0.78 [0.74, 0.83]	0.23	–	–
E6-SOLVER	euler_4_seq	256	2.472	2.186	1.21 [0.17, 2.25]	-1.84	–	–
E6-SOLVER	euler_8_seq	256	0.263	0.343	-0.47 [-1.59, 0.64]	-1.43	–	–
E6-SOLVER	euler_16_seq	256	0.162	0.189	0.20 [-1.17, 1.56]	0.01	–	–
E6-SOLVER	euler_32_seq	256	0.162	0.171	-0.39 [-1.95, 1.17]	-0.71	–	–
E6-SOLVER	euler_64_seq	256	0.166	0.166	1.03 [-0.68, 2.73]	0.99	–	–
E6-SOLVER	heun_4_seq	256	41.779	38.414	40.88 [31.69, 50.08]	0.53	–	–
E6-SOLVER	heun_16_seq	256	0.196	0.159	0.72 [-1.04, 2.49]	0.63	–	–
E6-SOLVER	euler_4_direct_vs_exact	256	80.997	78.910	79.14 [75.66, 82.61]	0.13	–	–
E6-SOLVER	euler_16_direct_vs_exact	256	9.461	8.699	9.56 [7.50, 11.61]	0.82	–	–
E6-SOLVER	euler_64_direct_vs_exact	256	0.894	0.699	0.85 [-0.43, 2.13]	0.23	–	–
E7-CFG	cfg_0.5_seq	256	30.620	30.792	31.16 [20.20, 42.12]	0.07	–	–
E7-CFG	cfg_1_seq	256	155.058	155.270	160.18 [117.73, 202.64]	0.23	–	–
E7-CFG	cfg_2_seq	256	676.600	676.538	674.62 [513.78, 835.45]	-0.02	–	–
E7-CFG	cfg_4_seq	256	1908.864	1907.677	1893.88 [1509.16, 2278.60]	-0.07	–	–
E7-CFG	cfg_1_direct_vs_exact	256	5.760	5.544	5.71 [2.80, 8.63]	0.11	–	–
E7-CFG	cfg_4_direct_vs_exact	256	76.689	76.409	84.87 [60.90, 108.84]	0.69	–	–
E5-MC	null_m32	32	0.000	0.000	5.25 [-4.72, 15.22]	1.03	-0.000	0.80
E5-MC	null_m64	64	0.000	0.000	0.24 [-4.49, 4.98]	0.10	-0.000	-0.52
E5-MC	null_m128	128	0.000	0.000	0.61 [-1.51, 2.74]	0.56	-0.000	1.03
E5-MC	null_m512	512	0.000	0.000	-0.11 [-0.65, 0.43]	-0.40	-0.000	-0.10
E5-MC	mean_shift_0.1_m32	32	5.646	5.646	14.98 [3.86, 26.10]	1.64	3.284	5.72
E5-MC	mean_shift_0.1_m64	64	5.646	5.646	8.00 [2.71, 13.29]	0.87	3.284	6.06
E5-MC	mean_shift_0.1_m128	128	5.646	5.646	6.11 [3.08, 9.13]	0.30	3.284	4.25
E5-MC	mean_shift_0.1_m512	512	5.646	5.646	5.14 [4.12, 6.17]	-0.96	3.284	2.92

D. Normal Reference Check for the Studentized Contrast

Table 4 gives, for each v1 null cell, the sign-flip distribution of the studentized mean of d_i given the observed $|d_i|$; it is the basis of the third audit finding in Section I.

Table 4. Sign-flip distribution of the studentized mean of d_i given the observed $|d_i|$ (4999 flips) for the seven v1 null cells.

Exp.	Condition	skew(d)	kurt(d)	t_{obs}	sd of t^*	$q_{0.95}(t^*)$
E1-NULL	`exact_sequential`	0.46	5.02	2.00	0.985	1.612
E4-MARKOV	`exact_sequential`	0.30	4.28	2.16	1.014	1.644
E4-MARKOV	`chain_drop_x`	-0.13	5.00	-0.40	1.014	1.634
E5-MC	`null_m32`	-0.78	7.58	1.03	1.005	1.686
E5-MC	`null_m64`	-0.04	5.58	0.10	1.000	1.641
E5-MC	`null_m128`	0.25	4.47	0.56	1.002	1.620
E5-MC	`null_m512`	0.08	5.87	-0.40	0.995	1.637

E. Numerical Components: Closed Form Versus Observation

Table 5. Numerical components with exact scores ($N = 100$, $M = 256$, Test S). D_{pop} is the closed-form population gap between the direct and sequential sampler laws, and $\mathbb{E}[d]$ is the closed-form expectation of the v1 contrast. All values $\times 10^3$, standardized.

Sampler (both branches)	$D_{\text{pop}} \times 10^3$	$\mathbb{E}[d] \times 10^3$	$\bar{d} \times 10^3$ [95% CI]	p_T
Euler, 2 steps	0.751	0.778	0.78 [0.74, 0.83]	0.001
Euler, 4 steps	2.472	2.186	1.21 [0.17, 2.25]	0.014
Euler, 8 steps	0.263	0.343	-0.47 [-1.59, 0.64]	0.804
Euler, 64 steps	0.166	0.166	1.03 [-0.68, 2.73]	0.121
Heun, 4 steps	41.779	38.414	40.88 [31.69, 50.08]	0.001
Heun, 16 steps	0.196	0.159	0.72 [-1.04, 2.49]	0.206
Euler 64, $w = 0.5$	30.620	30.792	31.16 [20.20, 42.12]	0.001
Euler 64, $w = 1$	155.058	155.270	160.18 [117.73, 202.64]	0.001
Euler 64, $w = 4$	1908.864	1907.677	1893.88 [1509.16, 2278.60]	0.001
Exact flow, $N \to \infty$	0.173	–	NOT RUN	–

F. Small Exact Verification (P5-E8-EXACT)

Table 6. P5-E8-EXACT, Test P with exact enumeration of the product permutation group ($N = 3$ examples, $M = 3$, $|G| = 20^3$), nominal $\alpha = 0.05$, one-sided 95% Clopper–Pearson bounds on the rejection rate over independent datasets. "Excess flagged" means the lower bound exceeds α; "powered" means the lower bound exceeds α under an alternative. Two negative-control rules are evaluated with the same criteria.

Cell	Role	rejections	1-s. 95% lower	1-s. 95% upper	Verdict
Exact sequential, target	null	9/200	0.024	0.077	no excess flagged
Corr. scale -1, target	null	4/100	0.014	0.089	no excess flagged
Exact sequential, joint	null	1/50	0.001	0.091	no excess flagged
Mean shift 2τ, target	alternative	42/50	0.730	0.918	powered
Variance ratio 4, target	alternative	5/50	0.040	0.199	not powered
Corr. scale -1, joint	alternative	4/50	0.028	0.174	not powered
Collapse intermediate, target	alternative	2/50	0.007	0.121	not powered
Always-accept rule	null / alt.	0/200	–	–	no excess flagged / not powered
Always-reject rule	null	200/200	–	–	excess flagged

1. **Conditional size.** For all 55 small datasets, including tied data from collapsed samplers, full enumeration confirms that the fraction of allocations with $p \leq \alpha$ is at most α for $\alpha \in \{0.01, 0.05, 0.1, 0.2, 0.5\}$.

2. **Monte Carlo agreement.** For 10 datasets, the Monte Carlo p-value with $B = 1999$ agrees with the exact one within tolerance and is never below $1/(B + 1)$; the smallest observed is 0.0025.

3. **Size and power (Table 6).** The exact null rejects 9/200 (one-sided bounds [0.024, 0.077]), and the target-level correlation null and the joint-level null show no excess. A 2τ mean shift is powered (42/50). Variance, correlation and collapse alternatives are *not* powered at this tiny size, as recorded without adjustment.

4. **Negative controls.** The always-accept rule fails the power criterion, and the always-reject rule is flagged for excess.

G. Quality Metrics in the Calibration Run

Table 7 lists the quality metrics of every calibration cell (Section I); the both-branch collapse cells pass the compatibility test and fail these metrics in every replicate.

Table 7. Quality metrics in the calibration run: mean CRPS excess over the true conditional ($\times 10^3$) for each branch, oracle sd ratio of the sequential branch, and the share of replicates whose 95% CI for the sequential CRPS excess lies above 0.

Cell	$\overline{\Delta\text{CRPS}}$ seq.	$\overline{\Delta\text{CRPS}}$ direct	sd ratio seq.	reps with CI > 0
Exact sequential	0.39	0.03	0.996	3%
Exact sequential (joint)	-1.00	0.79	0.996	3%
Corr. scale -1	0.24	0.57	0.996	3%
Chain $z \mid y$ (Markov)	-0.47	-0.24	0.997	3%
Mean shift 0.1τ	3.90	0.19	0.995	12%
Variance ratio 1.25	1.61	-0.65	1.113	14%
Collapse, intermediate only	12.20	0.13	0.732	44%
Corr. scale -1 (joint)	1.52	1.79	0.995	6%
Collapse, both branches	235.32	235.32	0.000	100%
Collapse, both branches (joint)	235.11	235.11	0.000	100%

H. v1 Controls with Test S

These are the first checker's pre-registered engineering controls on the Gaussian probe. They used Test S with $N = 200$ and $M = 256$. The single exact-null test rejected. That failure is discussed in Section I.

Table 8. Controls on the non-Markov probe (last row: Markov), Test S, $N = 200$ examples, $M = 256$. $\bar{d}$ and $\bar{d}_J$ are the mean target- and joint-level contrasts $\times 10^3$. ΔCRPS is the mean paired difference in CRPS between the compared and the reference branch $\times 10^3$; † marks the difference against the exact direct sampler, re-aggregated from raw. The sd ratio is the oracle sample sd over the true conditional sd. Every pre-registered detect / no-detect expectation in this table was met except for the exact null in the first row. The exact null on the Markov joint (not shown, $p = 0.015$) also rejected; it reused the same noise (Section I). Rows for 0.05τ and variance ratio 1.1 were pre-registered as descriptive.

Condition	$\bar{d} \times 10^3$ [95% CI]	p_T	$\bar{d}_J \times 10^3$	p_J	$\Delta\text{CRPS} \times 10^3$ [95% CI]	sd ratio
Exact sequential (null)	1.23 [0.03, 2.43]	0.026	0.64	0.057	3.2 [-5.2, 11.6]	1.00
Mean shift 0.05τ	2.03 [0.54, 3.53]	0.006	1.12	0.020	4.5 [-4.6, 13.7]	1.00
Mean shift 0.1τ	4.39 [2.80, 5.98]	0.001	3.46	0.001	7.2 [-2.8, 17.1]	1.00
Mean shift 0.5τ	136.98 [130.19, 143.78]	0.001	81.13	0.001	61.8 [21.2, 102.5]	1.01
Variance ratio 0.8	2.13 [1.00, 3.25]	0.002	2.74	0.001	6.1 [-1.2, 13.4]	0.89
Variance ratio 1.1	0.54 [-0.59, 1.66]	0.162	0.97	0.017	1.2 [-6.0, 8.3]	1.05
Variance ratio 1.5	13.31 [11.41, 15.21]	0.001	8.46	0.001	12.3 [2.3, 22.4]	1.22
Corr. scale -1 (target exact)	0.07 [-0.98, 1.12]	0.463	66.25	0.001	0.7 [-7.4, 8.8]	1.00
Corr. scale 0.5 (target exact)	0.36 [-0.76, 1.49]	0.278	6.62	0.001	1.4 [-6.6, 9.4]	1.00
Collapse, both branches	0.00 [0.00, 0.00]	1.000	0.00	1.000	239.2 [205.3, 273.0]†	0.00
Collapse, intermediate only	22.75 [20.97, 24.52]	0.001	111.60	0.001	17.4 [5.5, 29.4]	0.73
Observed intermediate	320.78 [269.99, 371.57]	0.001	–	–	-179.7 [-228.5, -130.8]	0.73
Chain $z \mid y$ (non-Markov)	39.28 [31.13, 47.44]	0.001	23.41	0.001	22.6 [0.4, 44.7]	0.93
Chain $z \mid y$ (Markov)	-0.18 [-1.08, 0.71]	0.667	0.19	0.291	-1.5 [-8.5, 5.4]	1.00

Table 8 summarizes the pre-registered controls; these are engineering checks of the checker, not scientific findings. Mean shifts of at least 0.1τ and variance ratios of 0.8, 1.25 and 1.5 are detected at the target level. A 0.05τ shift is also detected ($p = 0.006$; descriptive). A variance ratio of 1.1 is not detected at the target level ($p = 0.162$) but is detected at the joint level. The correlation breaks of Remark 3.3 leave the target marginal exact. They are not detected at the target level ($p = 0.278$ to 0.623) and are detected at the joint level ($p = 0.001$). Collapsing both branches gives $\bar{d} = 0$ exactly and $p = 1$, so the consistency check passes. Its CRPS is worse than the exact sampler's by 0.239 [0.205, 0.273] (0.815 vs. 0.575, 41.6% against the theoretical 41.4%), and its paired-index MSE is 0 against 2.02 for exact samplers. Collapsing only the intermediate is detected ($\bar{d} = 0.0227$) and yields sd ratio 0.73; the invalid paired-index MSE "improves" to 1.54. Feeding the observed intermediate is detected, and its CRPS is *better* than the direct branch's by -0.180 [-0.229, -0.131], the leakage predicted by Proposition 3.5. Chain generation is detected under the non-Markov joint and not under the Markov joint.

I. Calibration Development Record

Table 9. Calibration of Test P (single run; moved here from the main text). Rejection counts over independent replicates, with the one-sided 95% bounds used by the gate (together an equal-tailed two-sided 90% interval) and the two-sided 95% Clopper–Pearson interval.

Cell	family	rejections	1-s. 95% bounds	2-s. 95% CP	outcome
Exact sequential	null validity	16/400	0.025, 0.060	[0.023, 0.064]	within tolerance, no excess
Exact sequential (joint)	null validity	5/100	0.020, 0.102	[0.016, 0.113]	no excess
Corr. scale -1	null validity	5/100	0.020, 0.102	[0.016, 0.113]	no excess
Chain $z \mid y$ (Markov)	null validity	4/100	0.014, 0.089	[0.011, 0.099]	no excess
Mean shift 0.1τ	power	45/50	0.801, 0.960	[0.782, 0.967]	POWERED
Variance ratio 1.25	power	26/50	0.395, 0.643	[0.374, 0.663]	POWERED
Collapse, intermediate only	power	50/50	0.942, 1.000	[0.929, 1.000]	POWERED
Corr. scale -1 (joint)	power	50/50	0.942, 1.000	[0.929, 1.000]	POWERED
Collapse, both branches	quality control	0/10	0.000, 0.259	[0.000, 0.308]	not rejected; quality flagged
Collapse, both branches (joint)	quality control	0/10	0.000, 0.259	[0.000, 0.308]	not rejected; quality flagged

This appendix records the route to the calibrated test of Section 5.3, including both failures. Statistical wording follows the errata in Section N.

The failure. The v1 pre-registration required that the exact null not be detected and that a 20-replicate false-positive rate be compatible with 0.05. The single null test rejected at $p = 0.026$ (joint level $p = 0.057$), so the gate *failed*, and by our stop rule the v1 checker was not cleared for use on models. We keep this verdict. The replicates rejected 2/20 times at the target level (Clopper–Pearson 95% [0.012, 0.317]) and 1/20 at the joint level. The replicate z statistics had standard deviation 1.33 (χ^2 95% CI [1.01, 1.95]).

The audit. Three findings separate the gate from the estimator.

1. The gate was mis-specified. An exact α-level test rejects a single null with probability α. "Compatible with 0.05" over 20 replicates carries little information. The gate had no power requirement, so an always-accept rule would have passed it.

2. The v1 seeds omitted the joint and experiment names. The exact-null rejection on the Markov joint ($p = 0.015$) reused the same noise as the non-Markov one and is not independent evidence.

3. Test S is exact under Assumption 4.1 (Section 4.3), and the normal reference used for the reported z is adequate. Given the observed $|d_i|$ of each of the seven v1 null cells, the sign-flip distribution of the studentized mean has standard deviation 0.985–1.014 and 95% quantile 1.61–1.69 (normal: 1.645), although d_i has kurtosis 4.3–7.6 (Section D).

None of this shows that the checker is calibrated. It shows that the evidence does not demonstrate invalidity, and that the gate could not have established validity.

The revised test, checked exactly. We replaced Test S by Test P, introduced v2 seeds that hash every design factor, and pre-registered a small exact verification (P5-E8-EXACT) and a separate realistic calibration (P5-E8-CALIB) before running either.

On the smallest designs the checks behave as they must (Section F). Full enumeration confirms conditional size $\leq \alpha$ for all 55 datasets, including ties. Monte Carlo and exact p-values agree and are never 0. The exact null rejects 9/200 times. A 2τ shift is powered, but the variance, correlation and collapse alternatives are not powered at $N = M = 3$. The always-accept and always-reject rules fail. These are implementation checks of a textbook principle at $|G| \leq 8000$, not a calibration of the checker at the size used in Section H.

Auditing the calibration contract before running it. The first realistic calibration design (P5-E8-CALIB v2) was never run. A closed-form audit of its null and alternative cells, and of its gate, found three problems.

1. Its tolerance rule, a one-sided Clopper–Pearson upper bound ≤ 0.10, would fail a test with null rejection rate 0.05 (the least favourable rate for a level-0.05 test) with probability 0.56 in a cell with $R = 100$ and 0.20 with $R = 200$. At that rate, a test would have passed all v2 null cells with probability only 0.121.

2. The seed key of the small exact verification omitted the test level, so its target- and joint-level cells shared data.
3. There was no quality-control family, so the check could not show that a compatibility null which is a quality failure is handled correctly.

Table 10 states what each cell changes, in closed form. Three alternatives change the target law, the projected joint law and the distance to the truth. The correlation flip changes only the joint, so it is a null for the target test and an alternative for the joint test. Both-branch collapse changes neither branch law relative to the other, so it is a compatibility null, yet it is far from the truth.

The corrected design (v3) was fixed before any calibration result. It uses $N = 200$, $M = 64$ and $B = 199$, with seeds that include the level, and three separate families.

- **Null validity:** four cells.
  - The primary cell, exact sequential at the target level, has $R = 400$ and a tolerance rule. At a null rejection rate of 0.05 it fails with probability 0.019; a test of size 0.10 is caught with probability 0.964.
  - The other three cells have $R = 100$ and are checked only by excess flags at the Bonferroni level over the four cells. They catch a size of 0.10 with probability only 0.42.
- **Power:** four fixed alternatives, $R = 50$ each, all of which must be powered.
- **Quality control:** both-branch collapse, reported outside the gate.

In advance, at null rejection rate 0.05 and power 0.3 in each alternative, the gate passes with probability 0.945.

Table 10. Contract of the calibration cells (closed form, standardized scale). "Changed" compares the sequential with the direct branch; "truth" compares the sequential branch with the true conditional. $\mathbb{E}\,\Delta$CRPS is the expected CRPS excess over the true conditional.

Condition	target law changed between branches?	proj. joint law changed between branches?	truth (target) seq. vs. truth	sd ratio	$\mathbb{E}\,\Delta\text{CRPS} \times 10^3$ seq. / direct
Exact sequential	no	no	no	1.000	0.00 / 0.00
Chain $z \mid y$ (Markov)	no	no	no	1.000	0.00 / 0.00
Corr. scale -1	no	yes	no	1.000	0.00 / 0.00
Mean shift 0.1τ	yes	yes	yes	1.000	2.81 / 0.00
Variance ratio 1.25	yes	yes	yes	1.118	1.84 / 0.00
Collapse, intermediate only	yes	yes	yes	0.735	11.32 / 0.00
Collapse, both branches	no	no	yes	0.000	232.52 / 232.52

Calibration result (single run). The v3 calibration ran once, at the pre-registered commit, in 19.9 minutes of wall time on two processes (39.7 CPU-minutes). Table 9 lists every cell. The gate outcome was **PASS**: primary tolerance pass, null-family excess check pass, power family pass.

- **Null validity.** The primary null rejected 16/400 times (one-sided 95% lower and upper bounds 0.025 and 0.060, an equal-tailed two-sided 90% interval; two-sided 95% CP [0.023, 0.064]). The joint-level exact null, the target-level correlation null and the Markov chain null rejected 5/100, 5/100 and 4/100.
- **Power.** All four fixed alternatives were powered: 0.1τ shift 45/50, variance ratio 1.25 26/50, one-sided collapse 50/50, joint correlation flip 50/50.
- **Quality control.** The compatibility test did not reject both-branch collapse in any replicate, as it should not. The quality metrics flagged every replicate: CRPS excess over the truth about 0.235, sd ratio 0 (Table 7).
- **Negative controls.** The always-accept and always-reject rules fail the gate.

This is an engineering result on the Gaussian probe at this design. It shows no excess rejection at the tested nulls and adequate power against the tested alternatives. It does not establish level 0.05; the level guarantee rests on the permutation argument under Assumption 4.1. It does not show that the checker is calibrated for learned generators, whose samples need not satisfy the same assumptions. It also leaves the v1 record unchanged: the v1 gate failed, and the v1 checker was never cleared.

Consistency and quality answer different questions. Per-replicate CRPS excess over the true conditional is reported in Table 7 (Section G). Its per-replicate interval excludes 0 in 12% of replicates for the 0.1τ shift, and in 14% for the variance alternative. Test P rejected 90% and 52% of those replicates. The two are detection rates for *different* nulls, so neither number is a power comparison for the other's hypothesis.

- The CRPS interval asks whether the sequential branch is worse than the truth, using one held-out observation per example.
- Test P asks whether the two branches have the same law.

Both-branch collapse shows the converse case: it passes Test P and fails the quality check in every replicate.

J. Mixture Probe and Learner Design

Source. The source is a mixture of three components with weights $(0.5, 0.3, 0.2)$ and means $(-1, -1, 0)$, $(1, 1.5, 1)$ and $(0, 0.5, -1.5)$.

- Component standard deviations: $(0.6, 0.8, 0.5)$, $(0.5, 0.6, 0.7)$ and $(0.8, 0.5, 0.6)$.
- Component correlation matrices, as $(\rho_{xy}, \rho_{xz}, \rho_{yz})$: $(0.5, 0.2, 0.4)$, $(-0.3, 0.4, 0.2)$ and $(0.2, -0.5, 0.3)$.

All three covariances were checked positive definite before any data were drawn, and the parameters were not tuned.

Splits and seeds. The splits are 2000 train, 200 dev and 200 test draws. Data, model and sampling seeds use separate hash keys.

Fitted control. EM with $K = 2$, 500 maximum iterations, tolerance 10^{-8} and covariance regularization 10^{-6}. It is initialized by k-means++ seeding with the model seed. All marginals and conditionals are computed from the fitted density.

Learners.

- **Loss:** linear-path conditional flow matching.
- **Independent arm:** four MLPs (width 64, depth 3), one per pattern $z \mid x$, $y \mid x$, $z \mid x, y$ and $y, z \mid x$, with 5000 updates each.
- **Shared arm:** one MLP (width 128, depth 3) over (x, y, z) whose input marks each coordinate as observed, target or absent, with 20000 updates and uniform pattern sampling.
- **Budgets:** both arms get 20000 updates in total (4×5000 for the independent arm) and equal expected pattern exposure. Parameter counts are close (34757 and 34819), but each shared update runs a network about four times larger, so the arms are not equal-compute.
- **Optimizer:** Adam, learning rate 10^{-3}, batch size 256.
- **Solver:** Euler from $\mathcal{N}(0, I)$ with CFG off. The fine level (128) is primary, fixed in advance; the coarse level (32) is a sensitivity check that shares the same noise draws. The per-example difference of the two levels' gap estimates is reported with its paired 95% CI; an arm is flagged solver-sensitive when that CI excludes 0. Agreement does not prove convergence. The held-out flow-matching loss at 90% and 100% of training (fixed noise) is a diagnostic of optimization progress only. The replicates of Section 5.6 use 128 steps as primary and 256 and 512 steps as pre-fixed sensitivity endpoints on the same initial noise.
- **Checkpoint and comparison:** the final iterate, with no selection. The primary comparison is the paired per-example difference of the unbiased gap estimates of the two arms, over the same test examples.
- **Pre-execution fixes:** the shared input gained a target mask, and each independent network got its own optimizer.
- **Execution:** CPU only, one thread, torch 2.14.1+cpu; the adapter's fixtures (roles, leakage, path direction, optimizer isolation) were executed before training; the run contract was amended twice before training and not afterwards.

K. Development Pilot: Full Record

This is the single-seed pilot of Section 5.5 as reported before the replicates were run; the numbers are unchanged. Architecture, update budget, exposure and solver levels are in Section J; the pilot used 128 (primary) and 32 steps.

Table 11. Neural flow learners on the mixture probe (test split, Test P, $N = 200$, $M = 64$, $B = 199$). $\widehat{\Delta}_T$ and $\widehat{\Delta}_J^U$ are mean unbiased energy distances between the direct and sequential branches; $\widehat{D}_U(\text{direct}, \text{truth})$ is the unbiased sample energy distance between the 64 direct samples and the true $q(z \mid x)$ (closed-form cross and truth terms), which may be negative and is not the closed-form population distance of Table 1. All on the standardized scale; intervals are normal 95% CIs over examples.

Arm	steps	p_T	p_J^U	$\widehat{\Delta}_T \times 10^3$ [95% CI]	$\widehat{\Delta}_J^U \times 10^3$ [95% CI]	$\widehat{D}_U(\text{direct}, \text{truth}) \times 10^3$ [95% CI]
Independent conditional FM	128 (primary)	0.005	0.005	11.90 [7.19, 16.61]	11.83 [7.01, 16.65]	1.91 [0.02, 3.81]
Independent conditional FM	32	0.005	0.005	10.80 [6.26, 15.33]	11.27 [6.63, 15.91]	2.19 [0.34, 4.05]
Shared conditional net	128 (primary)	0.425	0.035	0.24 [-2.75, 3.24]	2.52 [0.28, 4.76]	5.51 [3.52, 7.51]
Shared conditional net	32	0.295	0.010	0.59 [-2.35, 3.54]	2.77 [0.59, 4.96]	7.00 [4.93, 9.07]

Consistency. At the primary endpoint the independent arm gave $p = 0.005$ and the shared arm $p = 0.425$ (Holm 0.010 and 0.425), with gap estimates $11.90[7.19, 16.61]$ and $0.24[-2.75, 3.24]$ ($\times 10^{-3}$). At 32 steps the p-values were 0.005 and 0.295; the paired solver differences of the gap estimates were $1.10[0.79, 1.41]$ and $-0.35[-0.64, -0.06]$ ($\times 10^{-3}$), flagged solver-sensitive: yes and yes. The paired between-arm difference $\widehat{\Delta}_T(\text{shared}) - \widehat{\Delta}_T(\text{independent})$ over the same examples is $-11.65[-16.93, -6.38]$ ($\times 10^{-3}$; percentile bootstrap over examples, 2000 resamples: $[-16.86, -6.30]$). The independent arm is flagged at the primary endpoint and the shared arm is not; by itself that would not show that the arms differ, and a larger p-value does not rank an arm as more coherent. The paired difference does speak to it: it favours the shared arm, and the same holds at 32 steps ($-10.20[-15.32, -5.09]$). The shared arm's projected-joint test, a secondary endpoint, gave $p = 0.035$ with gap $2.52[0.28, 4.76]$ ($\times 10^{-3}$): a smaller target gap did not come with an undetected joint gap, the direction of Proposition 2 of Klötergens et al. (2026) once more, and a reminder that parameter sharing defines no joint. All four solver flags are set. The gap estimates move by 0.3 to 1.1×10^{-3} between 32 and 128 steps, an order of magnitude below the between-arm difference but detectable under shared noise, so by the pre-registered rule the numerical and learned components are not separated at that level.

Correctness and cost. The direct branches lie at $\widehat{D}_U = 1.91[0.02, 3.81]$ and $5.51[3.52, 7.51]$ ($\times 10^{-3}$) from the true conditional, with spread ratios $0.980[0.965, 0.995]$ and $0.932[0.919, 0.946]$. The two arms are mirror images: the independent arm's direct branch is close to the truth while its sequential route is not ($9.84[6.95, 12.74]$), so its gap is error accumulated along the two-stage route; the shared arm's direct branch is the less accurate one (spread ratio $0.932[0.919, 0.946]$, mean shift $-0.045[-0.061, -0.028]$ true standard deviations) while its sequential route is at $2.39[0.40, 4.38]$. The shared arm's smaller measured target gap, which is not consistency (its joint endpoint is flagged), came with worse direct quality here; whether that trade-off holds beyond one seed and one probe is not known. Training took 32 and 47 CPU-s; sampling for the test split 14 and 35 CPU-s. The dev-loss diagnostic decreased by at most -0.16% and 1.04% over the last tenth of training (flagged: no and no). No density is available for these arms, so no likelihood is reported; and sharing parameters does not make the shared arm's conditionals those of one normalized joint, whatever its gap. One data seed, one model seed per arm, one synthetic probe: nothing here concerns real multimodal generators. A consistency regularizer, and any real multimodal data, are outside this paper ([TODO: P5-REAL-02: real paired multimodal data]).

Replicates: per-replicate results. Table 13 lists the eight arms of Section 5.6 at the primary solver, and Table 12 the paired between-arm differences at all three step counts.

Table 12. Paired between-arm difference at the target level at all three step counts ($\times 10^3$): per-replicate means with pointwise 95% CIs, $\widehat{\Theta}_h$ with its normal CI over examples, and the example bootstrap (2000 resamples; replicates, arms and step counts of an example carried together).

Quantity ($\times 10^3$)	$h = 128$ (primary)	$h = 256$	$h = 512$
$\bar{d}_{R1}(h)$ [pointwise 95% CI]	-11.18 [-17.70, -4.66]	-11.13 [-17.65, -4.60]	-11.10 [-17.63, -4.58]
$\bar{d}_{R2}(h)$ [pointwise 95% CI]	-22.81 [-29.58, -16.04]	-22.65 [-29.43, -15.88]	-22.57 [-29.35, -15.80]
$\bar{d}_{R3}(h)$ [pointwise 95% CI]	-8.15 [-13.11, -3.19]	-8.19 [-13.18, -3.21]	-8.21 [-13.22, -3.21]
$\bar{d}_{R4}(h)$ [pointwise 95% CI]	3.35 [-1.98, 8.67]	3.32 [-2.02, 8.65]	3.30 [-2.04, 8.64]
$\widehat{\Theta}_h$ [normal 95% CI over examples]	-9.70 [-12.57, -6.83]	-9.66 [-12.54, -6.79]	-9.65 [-12.53, -6.77]
bootstrap 95% (examples; replicates bundled)	[-12.47, -6.89]	[-12.68, -6.84]	[-12.57, -6.82]

Table 13. Four training replicates on the new test set (Test P at 128 steps, $N = 200$, $M = 64$, $B = 199$). $\widehat{\Delta}_T$ is the unbiased gap estimate; $\widehat{D}_U$ is the unbiased sample energy distance of a branch to the true $q(z \mid x)$ (may be negative; the negative interval of the R_2 shared arm is an unconstrained approximate interval, Section K); sd ratios are the sample sd of a branch over the true conditional sd. Standardized scale; intervals are normal 95% CIs over examples.

Rep.	Arm	p_T	p_J^U	$\widehat{\Delta}_T \times 10^3$ [95% CI]	$\widehat{D}_U(\text{dir}, \text{truth}) \times 10^3$	$\widehat{D}_U(\text{seq}, \text{truth}) \times 10^3$	sd ratio dir/seq
R1	Independent	0.005	0.005	13.52 [8.03, 19.00]	4.63	10.47	0.918/0.949
R1	Shared	0.085	0.050	2.34 [-1.28, 5.97]	3.78	4.47	0.934/0.998
R2	Independent	0.005	0.005	19.44 [13.50, 25.38]	3.28	14.54	0.973/0.999
R2	Shared	0.950	0.010	-3.37 [-6.29, -0.45]	9.72	7.67	1.034/1.039
R3	Independent	0.005	0.005	6.23 [2.16, 10.30]	2.67	8.55	0.961/0.944
R3	Shared	0.900	0.110	-1.93 [-4.90, 1.05]	2.84	5.07	0.957/0.936
R4	Independent	0.230	0.005	1.27 [-2.64, 5.18]	2.41	2.19	0.986/0.977
R4	Shared	0.035	0.010	4.61 [0.81, 8.42]	4.71	12.95	0.964/0.913

Estimator check on stored samples. Because the R_2 shared arm's direct branch has a negative interval for $\widehat{D}_U(\text{direct}, \text{truth})$ in Table 13, the estimator was re-checked once before any further run, on four fixed examples (indices 0, 50, 100, 150, chosen by position, not by sign) of all eight replicate arms at 128 steps: a brute-force double sum with the within-sample denominators $M(M-1)$ and the diagonal excluded, the closed-form cross and truth terms, and the standardization were recomputed from the stored samples and agree with the recorded values to within the storage precision ($\max|\text{diff}| < 10^{-7}$); the per-example values reproduce the recorded interval exactly; direct and sequential draws use distinct keys and a fresh y per z sample. No estimator or leakage defect was found. The interval is an unconstrained normal approximation over examples (140 of 200 per-example values are negative), and no negative population distance is claimed.

Replicates: solver sensitivity and cost. Tables 14 and 15 give the paired step-count differences and the cost of the eight arms.

Table 14. Solver sensitivity of the four replicates (target level, test set): paired per-example differences of the gap estimate and of the direct branch's distance to the truth between step counts, with shared initial noise ($\times 10^3$, normal 95% CIs over examples). Agreement is not a convergence proof.

Rep.	Arm	$\widehat{\Delta}_T(256) - \widehat{\Delta}_T(128)$	$\widehat{\Delta}_T(512) - \widehat{\Delta}_T(128)$	$\widehat{D}_U(512) - \widehat{D}_U(128)$
R1	Independent	-0.12 [-0.19, -0.06]	-0.18 [-0.28, -0.08]	-0.31 [-0.36, -0.25]
R1	Shared	-0.07 [-0.12, -0.02]	-0.11 [-0.18, -0.03]	-0.24 [-0.30, -0.18]
R2	Independent	-0.14 [-0.21, -0.07]	-0.21 [-0.32, -0.11]	-0.13 [-0.19, -0.07]
R2	Shared	0.01 [-0.03, 0.06]	0.02 [-0.04, 0.09]	0.11 [0.04, 0.17]
R3	Independent	0.04 [-0.01, 0.08]	0.06 [-0.01, 0.12]	-0.06 [-0.12, -0.01]
R3	Shared	0.00 [-0.04, 0.04]	-0.01 [-0.07, 0.06]	-0.08 [-0.14, -0.02]
R4	Independent	-0.05 [-0.11, 0.01]	-0.08 [-0.16, 0.01]	0.03 [-0.03, 0.09]
R4	Shared	-0.08 [-0.13, -0.03]	-0.12 [-0.20, -0.04]	-0.08 [-0.15, -0.02]

Table 15. Cost and diagnostics of the four replicates (one thread). The dev-loss decrease is the relative decrease of the held-out flow-matching loss between 90% and 100% of the updates (diagnostic only; two arms exceed the 2% flag).

Rep.	Arm	params	train CPU-s	test sampling CPU-s (128/256/512)	dev-loss decrease (max)
R1	Independent	34757	26	12/23/44	1.48%
R1	Shared	34819	37	25/48/96	2.01%
R2	Independent	34757	28	13/24/45	1.63%
R2	Shared	34819	39	26/48/96	0.05%
R3	Independent	34757	27	13/24/47	3.22%
R3	Shared	34819	39	27/50/100	1.10%
R4	Independent	34757	29	13/25/46	1.80%
R4	Shared	34819	40	25/50/97	1.39%

Cost and diagnostics of the neural arms. Table 16 is the pilot's cost record.

Claims table. Table 17 separates the three claims of Section 3.

Decision table. Table 18 summarizes what the mixture-probe runs decide.

Table 16. Cost and diagnostics of the neural arms. Train CPU-s and test sampling CPU-s are for one thread. "Dev-loss decrease" is the relative decrease of the held-out flow-matching loss (fixed noise) between 90% and 100% of the updates, a diagnostic of optimization progress only. "Solver paired" is the mean per-example difference $\widehat{\Delta}_T(128) - \widehat{\Delta}_T(32)$ with its paired 95% CI; agreement does not prove convergence.

Arm	params	updates	train CPU-s	sampling CPU-s (test)	dev-loss decrease 90→100% (max)	solver paired $\times 10^3$ [95% CI]
Independent conditional FM	34757	20000	32	14	-0.16%	1.10 [0.79, 1.41]
Shared conditional net	34819	20000	47	35	1.04%	-0.35 [-0.64, -0.06]

Table 17. Three claims that a consistency check is often taken to support, and what supports each here.

Claim	Evidence that would support it	Status in this paper
Non-detection of Δ_T, Δ_J^U	a calibrated, powered test that does not reject; reported with its design	synthetic probes, coherent controls and two neural flow arms (a pilot, four replicate initializations and an equal-CPU rerun)
Distributional correctness	proper scores or distances against a known truth	closed form on the probes
Coherent joint existence	all relations of the model, or a normalized density	by construction for controls only

L. Equal-Training-CPU Replicates (Supplementary Run)

The four replicates of Section 5.6 matched the number of parameter updates (20,000 per arm), so the shared network, which evaluates one width-128 network per update against four width-64 networks in round-robin, received roughly four times the per-update arithmetic. A single supplementary run, registered in full before it started (`configs/auto_run_r8.json`: stages, inputs and code hashes, seeds, test set digest, estimand, solver levels, stop conditions and caps), asked whether the measured pattern changes when the two arms receive the same process-CPU training time instead.

Design. Everything except the stopping rule is that of Section 5.6: the same training and dev rows, standardization, architectures and role masks, loss, Adam with learning rate 10^{-3}, batch 256, and the same integer seeds R_1–R_4 for initialization, minibatch stream and dev-loss noise. Each arm was retrained from step 0 and stopped after the first full update at which the process-CPU time charged to it reached 60.0 s, measured with `time.process_time()` in one single-thread process; for the independent arm this is the whole ensemble (four networks, balanced round-robin), for the shared arm the single network. Before each arm type a throwaway model of the same class was trained for 20 updates (seed 999) so that first-call costs were paid equally and were not charged. The final iterate is the model; there is no checkpoint selection. Overshoot beyond 60 s was at most 0.002 s (independent) and 0.003 s (shared). This is approximate OS-level matching, not exact FLOP matching; the update counts reached are in Table 21. A third test set of 200 examples was drawn from the same mixture (branch key `test_r8`; no x shared with the pilot's or the replicates' test sets) and is used by all four fit-pairs. Every draw is keyed by split, replicate, arm, pattern, example, branch, sample and purpose, with the step count absent, so Euler with 128 (primary) and 512 (sensitivity) steps integrates the same initial noise; a fresh y is drawn for every sequential z. Test P is run at 128 steps only ($M = 64$, $B = 199$); the eight target tests form one Holm family and the eight projected-joint tests a separate secondary family. The primary quantity is again $\widehat{\Theta}_{128}$, the mean over examples of the mean over replicates of $d_{ri}(128)$, with the normal interval over the 200 per-example replicate means and a cluster bootstrap over examples (2000 resamples carrying all replicates, arms and step counts of an example); the four replicate means are shown side by side and 4×200 values are never pooled.

Results. Status: COMPLETE. Table 19 gives the paired between-arm differences, Table 20 the per-arm tests and quality, and Table 21 the solver sensitivity, pattern exposure and overshoot. At 128 steps, 2 of the four independent arms and 1 of the four shared arms are flagged after Holm correction (raw p: independent $0.005, 0.015, 0.015, 0.005$; shared $0.005, 0.295, 0.505, 0.015$). The per-replicate means of $d(128)$ are $-9.15, -5.97, -4.52, -3.21$ ($\times 10^{-3}$); $\widehat{\Theta}_{128} = -5.71[-9.09, -2.34]$ (bootstrap $[-9.19, -2.54]$); sign reversal across replicates: no. By the replicate rule this is MIXED_REPLICATION: all four replicate means share the sign of $\widehat{\Theta}_{128}$, but three of the four pointwise intervals include 0 (only R_1's excludes it); the average is conditional on these four fits and is not a population effect. At the joint level, 0 of the four shared arms and 2 of the four independent arms are flagged in the secondary family; $\widehat{\Theta}^J_{128} = -6.82[-9.04, -4.59]$. The direct branches of the independent arms lie 3.8 to 13.3 ($\times 10^{-3}$) from the true $q(z \mid x)$ and their sequential routes 2.6 to 11.5; the shared arms' direct branches 3.5 to 7.0 and sequential routes 3.5 to 6.0. Going from 128 to 512 steps changes the gap estimate detectably in 1 of four independent and 0 of four shared arms (paired intervals), and $\widehat{\Theta}_{512} = -5.78[-9.17, -2.38]$; agreement between step counts is not a convergence proof, and where a paired difference is detectable the numerical and

Table 18. Decision table for the mixture-probe pilot.

Question	Status
Coherence vs. quality	controls: coherent by construction, not flagged at the primary endpoint; the fitted control's quality is clearly worse
Separate vs. shared learners	pilot (Table 11) and four replicates (Table 2): $\widehat{\Theta}_{128} = -9.70[-12.57, -6.83] \times 10^{-3}$ in favour of the shared arm, conditional on the four fits, with one sign reversal (MIXED_REPLICATION); the shared arm's joint endpoint is flagged in two of four
Equal training CPU (supplementary)	Table 19: $\widehat{\Theta}_{128} = -5.71[-9.09, -2.34] \times 10^{-3}$, same sign in all four pairs but detectable in one (MIXED_REPLICATION); the shared arm's direct branch is no longer worse; descriptive comparison only
Gap from solver refinement	not applicable to the controls (exact sampling); for the learners the paired step-count differences (32 vs. 128 in the pilot; 256 and 512 vs. 128 in the replicates) are reported and flagged when their CIs exclude 0
Non-detection vs. equality, correctness, global joint	for the controls equality and a global joint hold by construction, correctness fails for the fitted one; for the neural flow learners none of the three is established by a non-rejection; a global joint is not implied by parameter sharing

learned components remain unseparated. The run used 1646 CPU-s in one process (1657 CPU-s charged with the estimator check, fixtures and the timing smoke) and 1656 s of wall time.

Reading. Compared with the equal-update replicates of Section 5.6 ($\widehat{\Theta}_{128} = -9.70[-12.57, -6.83]$ with one reversal), the equal-CPU run gives a smaller average difference with the same sign in all four pairs and wider per-replicate intervals. The independent arms completed 8,075 to 8,206 updates per network (against 5,000) and the shared arms 23,058 to 23,730 (against 20,000); the independent arms' target gaps and direct-branch distances are larger than in Table 13 in two pairs, while the shared arms' direct branches (3.5 to 7.0) are no longer worse than the independent arms' (3.8 to 13.3), so the accuracy trade-off seen at equal update counts does not recur at equal CPU. At the joint level no shared arm is flagged after Holm correction (raw p: $0.015, 0.885, 0.050, 0.020$), against two of four in Section 5.6, while the independent arms' raw p are $0.005, 0.055, 0.010, 0.005$. At the joint level 3 of the eight paired 512-versus-128 differences are detectable (at most 0.05×10^{-3} in magnitude). The comparison with Section 5.6 is descriptive: the test sets differ, the design had been seen before this run was registered, and matching process CPU is not matching FLOPs or wall time on other hardware. Nothing here is a population statement over training runs, and non-detection of a shared arm is neither equality nor a coherent joint.

Table 19. Equal-training-CPU run: paired between-arm difference at the target level ($\times 10^3$), per-replicate means with pointwise 95% CIs, and $\widehat{\Theta}_h$ over examples (conditional on the four fits) with its normal interval and the example bootstrap.

Quantity ($\times 10^3$)	$h = 128$ (primary)	$h = 512$
$\bar{d}_{R1}(h)$ [pointwise 95% CI]	-9.15 [-17.21, -1.09]	-9.22 [-17.30, -1.14]
$\bar{d}_{R2}(h)$ [pointwise 95% CI]	-3.21 [-9.37, 2.94]	-3.30 [-9.48, 2.89]
$\bar{d}_{R3}(h)$ [pointwise 95% CI]	-4.52 [-11.13, 2.09]	-4.69 [-11.38, 2.00]
$\bar{d}_{R4}(h)$ [pointwise 95% CI]	-5.97 [-12.57, 0.63]	-5.90 [-12.52, 0.72]
$\widehat{\Theta}_h$ (normal)	-5.71 [-9.09, -2.34]	-5.78 [-9.17, -2.38]
$\widehat{\Theta}_h$ (cluster bootstrap)	[-9.19, -2.54]	[-9.26, -2.31]

Table 20. Equal-training-CPU run: the eight arms on the third test set (Test P at 128 steps, $N = 200$, $M = 64$, $B = 199$). Updates are the parameter updates completed within 60 process-CPU seconds; $\widehat{\Delta}_T$ is the unbiased gap estimate; $\widehat{D}_U$ is the unbiased sample energy distance of a branch to the true $q(z \mid x)$ (may be negative; unconstrained intervals, Section K); CPU-s is the training time charged. Standardized scale; intervals are normal 95% CIs over examples.

Rep.	Arm	updates	p_T	p_J^U	$\widehat{\Delta}_T \times 10^3$ [95% CI]	$\widehat{D}_U$(dir, truth) $\times 10^3$	$\widehat{D}_U$(seq, truth) $\times 10^3$	CPU-s
R1	Independent	32709	0.005	0.005	15.15 [9.02, 21.29]	13.28	11.48	60.0
R1	Shared	23730	0.005	0.015	6.00 [1.11, 10.90]	7.05	5.95	60.0
R2	Independent	32824	0.015	0.055	4.60 [0.25, 8.95]	3.80	5.69	60.0
R2	Shared	23535	0.295	0.885	1.39 [-3.69, 6.46]	4.23	4.99	60.0
R3	Independent	32331	0.015	0.010	4.63 [-0.74, 10.00]	6.73	4.50	60.0
R3	Shared	23259	0.505	0.050	0.11 [-3.62, 3.84]	3.69	3.53	60.0
R4	Independent	32301	0.005	0.005	10.17 [4.84, 15.50]	6.36	2.59	60.0
R4	Shared	23058	0.015	0.020	4.19 [0.59, 7.79]	3.53	5.66	60.0

M. Pre-Registration Record and Deviations

- **v1.** Hypotheses EH1–EH6, the statistic, $\alpha = 0.05$ and the stop rule were fixed before the test run.
  - EH2–EH6 matched their predictions. EH1 failed and is kept as failed.

Table 21. Equal-training-CPU run: paired step-count differences (512 minus 128, shared initial noise, $\times 10^3$, normal 95% CIs over examples) of the gap estimate and of the direct branch's distance to the truth; pattern exposure (updates per conditioning pattern); and the CPU overshoot beyond 60 s at the stopping update. Agreement between step counts is not a convergence proof.

Rep.	Arm	$\widehat{\Delta}_T(512) - \widehat{\Delta}_T(128)$	$\widehat{D}_U(512) - \widehat{D}_U(128)$	exposure (z|x / y|x / z|x,y / y,z|x)	overshoot s
R1	Independent	0.00 [-0.10, 0.10]	0.39 [0.31, 0.47]	8178/8177/8177/8177	0.001
R1	Shared	-0.07 [-0.16, 0.03]	0.22 [0.16, 0.28]	5935/5990/5854/5951	0.002
R2	Independent	0.01 [-0.07, 0.09]	0.02 [-0.06, 0.09]	8206/8206/8206/8206	0.002
R2	Shared	-0.07 [-0.15, 0.01]	0.22 [0.15, 0.29]	5887/5788/6028/5832	0.003
R3	Independent	0.19 [0.09, 0.29]	0.14 [0.07, 0.21]	8083/8083/8083/8082	0.002
R3	Shared	0.02 [-0.06, 0.09]	0.16 [0.10, 0.22]	5731/5915/5851/5762	0.002
R4	Independent	-0.06 [-0.15, 0.04]	-0.18 [-0.25, -0.12]	8076/8075/8075/8075	0.002
R4	Shared	0.02 [-0.08, 0.12]	-0.16 [-0.21, -0.10]	5811/5785/5746/5716	0.002

  - Post hoc we found: the mis-specified EH1 criterion; the seed flaw that shares noise across joints; and a rendering bug in a table script, which did not affect the raw data.
  - A reported-statistics error in our own audit notes (a miscounted number of cells) was corrected in the record.

- **P5-E8-EXACT** was fixed before running.
  - Components A and B passed their deterministic criteria.
  - In component C, the three pre-registered null cells showed no excess, and the pre-registered powered alternative was powered. The three "report" alternatives were not powered.
  - A development smoke run with a different seed preceded it.

- **P5-E8-CALIB, v2** was fixed before any E8 result, except its resource estimate, which it declared would be filled from the E8 timing smoke.
  - A pre-run contract audit showed that its gate would fail an exactly valid test with probability $1 - 0.121$.
  - The audit also found that the E8 seed key omitted the test level, and that v2 had no quality-control family.
  - v2 was never run and is preserved.

- **P5-E8-CALIB, v3** replaced v2 before any calibration result.
  - It has three families: null validity, power, and quality control.
  - Primary null: $R = 400$ with a tolerance criterion. The other null cells use Bonferroni excess flags.
  - Seeds include the test level.
  - The gate's operating characteristics were computed in advance: P(pass | valid, power 0.3) $= 0.945$.
  - A timing smoke with a different seed preceded the single run and is not reported.

- **Equal-training-CPU run** (Section L) was registered in full before it started: allowed stages, input and code hashes, the R7 seeds reused unchanged, the third test set and its digest, the 60 s process-CPU stop rule with warm-up, solver levels 128/512, the estimand, both Holm families, stop conditions and caps; the outcome label applies the replicate rule unchanged. A single run; nothing was selected by outcome. Deviations: none; the registered command ran once at the registered commit with matching code hashes, the test-set digest was verified, and all eight arms reached the budget with finite losses. No dev-loss undertraining diagnostic is defined for a CPU-budget stop, so none is reported.

- **Exploratory analyses.** The re-analysis in Sections C and D and the closed-form curves in Figure 1 were computed after the v1 results were seen.

N. Errata to Earlier Versions

1. **Size.** "Size exactly 0.05 for $B = 199$" is replaced by "level $\leq \alpha$". Exact size needs, in addition, that the permuted statistics have no ties almost surely, which is not guaranteed (Section 4.3).

2. **Intervals.** Pairs of one-sided 95% bounds were written as if they were one interval. Such a pair is an equal-tailed two-sided 90% interval; two-sided 95% intervals are now reported.

3. **CRPS vs. Test P.** Their detection rates concern different nulls and are not a power comparison.

4. **Scope of the calibration pass.** The Gaussian calibration pass is engineering evidence for one implementation and design. It is not an empirical guarantee on new data.

5. **Naming.** A network shared across conditionals is a shared conditional network, not a coherent joint.

O. Compute

All runs are CPU-only. The Gaussian-probe runs and the mixture controls are single-process standard-library Python; the calibration used two processes; the neural pilot and the four replicates used torch (2.14.1+cpu), one process, one thread.

- v1 test run: 80.9 s.
- P5-E8-EXACT: 1.1 s.
- Development, timing and analysis runs are listed in the supplementary compute ledger.
- P5-E8-CALIB v3: 19.9 minutes of wall time on two processes (39.7 CPU-minutes); 920 tasks, run once.
- A calibration timing smoke with a different seed is listed in the ledger and not reported.
- Mixture controls (P5-SYN-LEARN-01): 22.5 s. Neural pilot: 220 CPU-s for the run process (238 CPU-s charged with fixtures, smokes and preflight), 227 s wall. Four replicates: 1494 CPU-s (1500 CPU-s charged), 1517 s wall, one thread. Equal-training-CPU run (Section L): 1646 CPU-s (1657 CPU-s charged with the estimator check, fixtures and smoke), 1656 s wall, one thread. Timing smokes (seed 999) are in the ledger and not reported.

P. Reproducibility

The supplementary code regenerates every number in this paper:

- `run_first_run.py --split test` reproduces the v1 raw results.
- `run_e8_exact.py` reproduces P5-E8-EXACT.
- `reanalyse_v1.py` reproduces the exploratory re-analysis.
- `run_calib.py` reproduces P5-E8-CALIB v3; `run_syn_learn.py --split test` the mixture controls; `run_syn_learn_fm.py` the neural pilot, `run_syn_learn_fm_r7.py` the four replicates and `run_syn_learn_fm_r8.py` the equal-training-CPU run (CPU torch environment as recorded in the supplement; the CPU-budget stop makes the update counts of that run hardware-dependent, so its checkpoints are stored).
- `make_paper_assets.py` regenerates the tables, figure data and numeric macros from committed raw results.

Unit tests cover the linear algebra, the conditionals and perturbations, the sorted energy distance against brute force, the fair CRPS against its Gaussian closed form, the affine PF-ODE map against the sampler, the V/U equivalence and exact-enumeration super-uniformity of Test P, and the Clopper–Pearson bounds. They are numerical checks, not proofs.